

BORDER OPTICS — Research Paper

Satellite Verification of India's Vibrant Villages Programme: A Compositing-Window Robustness Assessment of Border-Village Development

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Satellite Verification of India's Vibrant Villages Programme: A Compositing-Window Robustness Assessment of Border-Village Development

Sakshi D. Maske Independent Geospatial Researcher

Abstract

Ask the Government of India if things have worked out for them with their flagship scheme for the Border Villages, the honest answer (on parliamentary record) is No, it has never been verified anywhere. The Vibrant Villages Programme (VVP-I) has approved around ₹4,800 crore in support of 2,967 villages in five Himalayan border states/union territories including 662 villages marked 'priority' for Phase-I of the programme since 2022-23. There is no impact assessment in any manner. This study utilizes two satellite proxies: Sentinel-2 built-up-index (NDBI) and VIIRS night-lights at 258 individually geocoded villages across three core states (Arunachal Pradesh, Sikkim, Uttarakhand), an illustrative sample of 7 villages in Himachal Pradesh, and compares them with a district-restricted non-VVP control group of 732 villages across the same 14 districts (deduplicated against the treated sample by coordinate proximity and by the full official priority-village name list, not just by exact name match against successfully-geocoded villages — see Section 3.7). Everything was calculated twice, with two defensible methods of compositing the same 2021-2025 satellite record, specifically to check whether the answer depends on that choice. On the headline test, it mostly doesn't, once both windows are extracted completely and on the same day as the control group: neither shows a significant treated-only built-up-area increase (full-year Wilcoxon $p = 1.000$; summer-matched, June-September, $p = 0.435831$, $n = 251$). That second number is the one that changed. An earlier version of this same summer-window test, run before all 31 Sikkim villages' monsoon-season imagery had backfilled into the Sentinel-2 archive, returned $p < 0.000001$ on a 154-village subsample missing Sikkim entirely — a result that looked like a clean, surviving signal until the complete

sample was actually available (see Development Log Entries 21-22 for the full account, including two extraction-script bugs found and fixed in the process of getting there). Against the control group, the story is the same: neither window's NDBI gap holds (full-year did coefficient = +0.0067, $p = 0.310$; summer = -0.0036, $p = 0.473$, now on the complete 251-treated/732-control sample), nor does the buffer-radius sweep (250m/500m/1km, all now pulled the same day, all null: $p = 0.1214/0.4358/0.8976$), nor the leave-one-district-out check (0 of 14 reruns significant, against 14 of 14 on the earlier, incomplete sample), nor a randomization-inference test that doesn't depend on cluster asymptotics ($p = 0.154$). Night-lights tells the same story on its own treated-only terms — no absolute change in either window (Wilcoxon $p = 0.050$ full-year, $p = 0.9999$ summer) — and, as of a fix to the control-group list that removed 23 villages that were physically or nominally duplicates of treated villages (Development Log Entries 23-25), also against the control group in both windows: a full-year gap that had briefly looked like this study's one surviving significant result is null on the corrected, contamination-free control list ($p = 0.309$), and fails the same leave-one-district-out and randomization-inference checks that had previously been cited as evidence it was holding up. This primary null has since been tested two further ways it had not yet faced: an exact wild-cluster bootstrap (16,384 sign-flip combinations, addressing this study's reliance on only 14 district clusters) leaves all four outcome/window DiD results non-significant ($p = 0.32$ - 0.50), and a genuine 2019-to-2021 parallel-pre-trends placebo test — something the original single-pre-period control design could not support — is clean in three of its four combinations, with the fourth (summer NDBI, $p = 0.064$) borderline and reported as such.

Built-up change also correlates with distance to the border/LAC (H3), in the opposite direction from what securitization theory predicts — summer NDBI ($\rho = 0.291$, $p = 0.0000026$) and full-year lights ($\rho = -0.252$, $p = 0.0000548$). Once described as a fresh, unstress-tested finding, this correlation has since been run through leave-one-district-out (14 of 14 reruns significant, same sign), randomization inference ($p < 0.001$), and a spatial-autocorrelation-calibrated permutation test that treats the 251 villages as the non-independent observations they actually are — and survives all three (spatially-corrected $p = 0.0295$ summer NDBI, $p = 0.0035$ full-year lights), making it this study's one real secondary finding rather than an open question.

One further check did not simply confirm the primary null and is reported as an open, unresolved tension rather than smoothed into it. Triangulating NDBI/lights against two satellite proxies found Sentinel-1 SAR backscatter (the only genuinely sensor-independent one) agreeing (null, both windows, both polarizations), but Dynamic World's machine-learned built-up probability disagreeing: a significant control-group DiD in both windows ($p = 0.0117$ full-year, $p = 0.0013$ summer) that is hard to dismiss, since Dynamic World also validates far more strongly against an independent, non-satellite ground truth (Google Open Buildings footprint counts, $\rho = 0.808$ vs. NDBI's $\rho = 0.399$) and correctly detected all three of this study's confirmed real-construction case studies (Kaho, Walong, Musai) where NDBI moved in the wrong direction at all three. A second check that initially looked like a second open tension — cross-checking the QA60 cloud mask against an SCL-band alternative, which first appeared to flip the summer NDBI result from null to significant — turned out on external review to be a coding bug (the SCL mask was letting snow pixels through as "clear"); fixed and rerun on the corrected 251-village sample, the two masks agree, both null ($p = 0.436$ QA60, $p = 0.288$ SCL). That is reported as a resolved bug caught by review, not as a second surviving tension.

Budget still doesn't track outcome in a way worth building a claim on: with all three core states now having valid summer data, Arunachal Pradesh's roughly tenfold larger sanctioned budget over

Uttarakhand's corresponds to a smaller mean NDBI change, and Sikkim's smallest budget of the three corresponds to a negative one — descriptive only, $n = 3$, and the correlation's own sign flips between the two compositing windows. None of this is the verdict this study originally reported, and that reversal is itself part of the finding: on its primary measure, once the extraction is complete, treated/control data pulled the same day, and the design stress-tested against small-cluster bias and a genuine pre-treatment placebo, no satellite-detectable effect of VVP-I survives — a real, reportable answer to a question Parliament's own record says was never asked. Triangulating against an independently ground-truth-validated proxy surfaced a genuine sensitivity in that primary measurement this study reports honestly rather than resolves, alongside a border-proximity pattern that, unlike it, has now been fully stress-tested and holds; a second apparent sensitivity, in the cloud-masking convention, turned out under review to be a bug rather than a real one.

Keywords: Vibrant Villages Programme, border development, securitization theory, satellite verification, NDBI, VIIRS night-lights, SAR backscatter, Dynamic World, building footprint validation, remote sensing, robustness testing, wild cluster bootstrap, spatial autocorrelation, cloud masking sensitivity, difference-in-differences.

1. Introduction

India's border villages have been part of the discourse of policy for long, as liabilities – they are found in remote areas, are sparsely populated, and prone to being lost out – towards a 'contested frontier'. That's the solution the Vibrant Villages Programme (VVP-I) has proposed and taken to the Union Cabinet for approval, which cleared the project in February 2023, turning 2,967 villages in Arunachal Pradesh, Sikkim, Uttarakhand, Himachal Pradesh and Ladakh into development priorities, not security afterthoughts. One thing is missing from that story, however, more than two years after sanction gone by, there has been no independent basis of determining whether the promised development actually occurred. This time, Parliament directly asked, has VVP made an impact? The reply was unambiguous: No impact assessment has been done (Lok Sabha Unstarred Question No. 508 of 3 February 2026, Ministry of Home Affairs).

This study is just that. Test—village by village—whether any physical development appears on the ground because of sanction; whether the size of a village correlates with any of the state's budgets; whether its location is more closely related to being near the border as opposed to the level of development required. Here is the place of securitization theory: If border infrastructure is a geopolitical sign, rather than a welfare policy measure, then results should be as correctly predicted as crucially as need is.

There are five research questions that emerge from this, namely, does expansion of built-up area happen at priority villages since sanction (RQ1); does the size of this expansion correspond to the budget allowed (RQ2); do the night-lights confirm the picture of built-up area changes (RQ3); is development concentrated along the border and not where it is most needed (RQ4); and is all of this growth linked to VVP-I or is it linked at the village level generally (RQ5)? Four hypotheses provide operationalizations of the questions. Time since the implementation of the policy makes a big difference on the measure of built-up area. This increase will not grow proportionally to budget: the increase will be relatively smaller in the states with relatively bigger budget. Villages closer to the border/LAC seem to transform more rapidly than those further away within the same State, as securitization theory would suggest. The test that identifies a programme effect beyond the general trend – or difference (H4) –

predicts that change will be significantly higher in VVP-I priority villages as compared to a district-restricted set of non-VVP villages within the same districts and the same period.

2. Literature Review

2.1 Securitization Theory and Border Development

By shifting the agenda from “ordinary” politics to securitisation, opportunities to unlock resources and “extraordinary” measures that wouldn’t be politically possible arise (Buzan, Wæver, & de Wilde, 1998). Spending on border development is an almost perfect fit into this picture: As spending on welfare, it is legitimate; as spending to build strategic links, it is also legitimate; and money is seldom separated in policy literature. Here, VVP-I is used as a test case of that theory: is geographic proximity to the border an even better indicator of measurable development than the conventional indicators of development — existing infrastructure that would be in place anywhere, for example —?

2.2 The Vibrant Villages Programme: Policy Context and the Accountability Gap

However, the paper trail in VVP-I’s scope lies not altogether in one place in the parliamentary replies. The complete village-wise annexure has been provided in the Rajya Sabha Hansard (Question No. 2321, Ministry of Home Affairs dated 9 August 2023). The 75 priority villages of Himachal Pradesh have been confirmed in Lok Sabha Question No. 2104 (2023) and in Rajya Sabha Question No. 401 (2025) but there is no list of the villages at the name level. Ladakh’s 35 sanctioned villages are officially confirmed as such in Lok Sabha Question No. 4360 (2025), but without measures published giving the names of the villages. And answering the question that comes to everyone’s mind when reading this article — Lok Sabha Unstarred Question No. 508 (3 Feb 2026) — Shri Baijayant Panda asked whether VVP’s effect has been assessed and the Ministry’s reply was straightforward — “No impact assessment has been done”. As in searching any of the governmental sources, there are none that present an independent, satellite-verified account of actual advances made over the predetermined scope. This study comes to fill that void.

2.3 Remote Sensing Approaches to Built-Up Area and Economic Activity Detection

Here two indices take the analytical weight, selected for various reasons. A widely-used proxy for built-up surface extent in multi-temporal comparison is the Normalized Difference Built-up Index derived from the reflectance in the short-wave infrared and near-infrared bands (Zha, Gao, & Ni, 2003). The source of radiance as measured by the VIIRS Day/Night Band is somewhat related, but quite different: the amount of electrification and economic activity, and it has excellent dynamic range and resolution compared to the older DMSP-OLS (Elvidge, Baugh, Zhizhin, Hsu, & Ghosh, 2017). This study uses two indices that work well with each other (NDBI versus vegetation phenology and snow; VIIRS versus cloud cover and sensor saturation at low radiance). If either one disagrees with the other, it doesn’t necessarily mean either is wrong — but where they agree, that carries more weight than either alone.

2.4 Compositing-Window Sensitivity in Multi-Temporal Satellite Analysis

Change detection for spectral data is difficult for high-relief terrain in certain ways: First, snow cover and cloud frequency vary both by season and elevation; and second, a choice of compositing that isn't elevation-sensitive can create or obscure the signal of true change. This study doesn't consider that sensitivity as an averageable noise. It's being tested directly, as a characteristic of the result itself, and also, which is more important, one that may be reversed between two enthusiastically defensible compositing windows — if either of them were a sole source of confirmation, then that would be the test, wouldn't you focus it on them?

3. Data and Methodology

3.1 Study Design

Only three States, namely Arunachal Pradesh, Sikkim, and Uttarakhand involving 251 villages alone which can be identified on the basis of the village-wise information officially available, are part of the core statistical sample. The 7 confirmed villages of Himachal Pradesh are summarized separately as an exemplar village study, rather than included as part of the core sample, and Ladakh is not included in any village-level study (Section 6.1) because of a documented data-availability gap. The difference between the villages in each group before and after treatment is compared against a difference-in-differences specification (H4) that includes district fixed effects — removing each district's own time-invariant baseline level, not a district-specific *trend*, since the model's post-period term is shared across every district and village alike — and a comparison to a district-restricted non-VVP control group of 732 villages in the 14 districts, to net out whatever common secular trend the wider region experienced over the same period, VVP-affiliated or not. Two additional checks take the translation of “2021-vs-2025” up a level: A third, independent time point (2023) follows; the extraction buffer is fixed at 500m as throughout, and is tested out again at 250m and 1km to determine if the change of the extraction buffer's magnitude changes the answer as well as the compositing window.

3.2 Data Sources

Variable	Source	Temporal Coverage
Village identification	State VVP-I portals; Rajya Sabha/Lok Sabha Q&A annexures	2023-2025
Village coordinates	OpenStreetMap Nominatim; ISRO Bhuvan Village Geocoding API	Current
Non-VVP control village identification	OpenStreetMap Overpass API, district-matched to treated sample	Current
Built-up index (NDBI)	Sentinel-2 SR Harmonized	2021, 2023, 2025
Night-lights radiance	VIIRS DNB monthly composites	2021, 2023, 2025
Border/LAC geometry	Natural Earth 10m Admin-0 Boundary Lines	Current
Budget/project	State-wise VVP-I sanction figures	2023

The village-level datasets first had to be compiled and geocoded.

3.3 Village-Level Dataset Compilation and Geocoding

Although the primary geocoding was done via Nominatim, Bhuvan was used as a Census-linked fallback — found necessary because Bhuvan doesn't filter by state (a test query for the Arunachal Pradesh village 'Kharman' initially returned an unrelated same-named village in Haryana), so manual district validation was applied throughout. All the results, both from sources, were manually validated in the districts before accepting and any village list was cross validated with the official aggregate figures first. Of the 559 villages attempted across the four states, 258 were successfully geocoded — 186 of 455 in Arunachal Pradesh, 31 of 46 in Sikkim, 34 of 51 in Uttarakhand, and 7 of 7 in Himachal Pradesh.

3.4 Satellite Change Detection

A buffer of 500m around each geocoded village was applied to collect mean NDBI and VIIRS night-lights radiance by using Google Earth Engine 2021 compared to 2025. The NDBI is derived from Sentinel-2 Level-2A Surface Reflectance bands B11 (SWIR1) and B8 (NIR), where cloud was masked using the QA60 band. Night-lights were derived from VIIRS Day/Night Band (NOAA/VIIRS/DNB/MONTHLY_V1/VCMSLCFG).

3.5 Compositing-Window Robustness Testing

Both of these measures were calculated twice - without and with the subtraction of an annual season matched measure (full composite, season matched), designed specifically to remove real change from seasonal artifact (snow in full composite, monsoon cloud in season matched composite). If there was no valid composite in a window, this village did not get zeroed out, it would be marked null.

3.6 Border-Proximity Analysis

The distance calculations were done in GeoPandas and the distance to only those segments of Natural Earth Admin-0 on Indian territory that are relevant to the village was computed. However, there is one proviso that applies to all the numbers that follow this: The line of actual control (LAC) between India and China here is contested, and there is no hard determinate agreed border line. Thus, all of the distances discussed in this study are only relative and comparative and in no way an authoritative delineation of the location of the border.

3.7 Non-VVP Control Group and Difference-in-Differences Design

A treated-only comparison can't rule out a clear alternative explanation: even in the absence of any real VVP-I effect, the same shift could have been caused by conditions shared by all these villages, irrespective of any treatment — anything from unrelated infrastructure spending, unrelated state initiatives or chance regrowth after a low year would do the same. These villages have been dynamically assembled from the same 14 districts as the sample villages using the OpenStreetMap Overpass API, excluding any village that is on a VVP-I priority list, and then processed through the same

set of nodes, 500m buffer, NDBI/VIIRS definitions and before/after windows. Results were reshaped to be a panel of two periods with a treatment indicator and then tested as

$\text{ndbi (or lights)} \sim \text{treatment} + \text{post} + \text{treatment} \times \text{post} + \text{district fixed effects}$

Using the same district level variable (14 clusters), standard errors were clustered. But the coefficient of interest is that of the treatment $\times$ post, which is the difference between the treated group's change and the control group's change in the same time frame, net of the district fixed effects component of any baseline growth trend that varies across districts. A parallel HC3 heteroskedasticity-robust specification without fixed effects is reported together with that as 14 clusters is below the 30–40+ recommended for cluster-robust asymptotics to be fully trustworthy. A fully parallel-pre-trends placebo (the sort of design the project's own earlier control-group research have run elsewhere using a multi-period pre-treatment panel) can not be run on this design. Here the satellite extraction in the control group only includes the same before/after pair that the treated sample has, thereby the level-balance check (Mann-Whitney U) is a weaker at best substitute for a baseline check (2021).

3.8 Multi-Year Trend Extraction

Two single years, 2021 and 2025, are clearly an open possibility how to construct or destroy an apparent change that actually has nothing to do with real development, but can be triggered by weather variation. A third time point was selected, namely 2023, about halfway between the two, extracted for the same 251-village core sample under both compositing windows, and yields one value per village, annually, not another single 'before' and 'after' delta. A per-village linear trend was originally fitted over the three years and the signed ranks Wilcoxon test was used to test the slope against zero, as is often done with this kind of time series model. An additional and alternative analysis was a village-fixed-effects panel regression ($\text{value} \sim \text{year}$, clustered SE by village) applied to the slopes of the linear trend model. The two year spans 2021–2023 and 2023–2025 were also tested independently to generate a maximum concentration with a change in one half of the window.

3.9 Buffer-Radius Robustness Testing

In the other tests in this study, the 500m buffer was used as a fixed methodological choice throughout, not tested against other buffers until now. The same extraction for the summer window was repeated at 250m and 1km for the same core sample, and a Wilcoxon signed-rank test was performed on each radius. A complication arose because the 250m and 1km extractions were run at a later date than the original 500m extraction — against the identical fixed 2021/2025 date ranges, but a Sentinel-2 archive that keeps backfilling scenes close to the present — so a raw village-count comparison across the three buffer radii is confounded by that archive-timing difference rather than isolating buffer radius alone. It will come down to making sure that the three radii are restricted to the subset of villages that have data for all three, fixing sample composition and leaving only buffer radius as it varies.

3.10 Statistical Testing

Prior to/after change, Wilcoxon signed-rank test of difference (Wilcoxon, 1945) – paired, non-parametric, appropriate because the sample was not normally distributed. Due to their robustness to non-linear monotonic relationships, Spearman's rank correlation (Spearman, 1904) was used for budget correlation (RQ2, descriptive

only – three States now have sufficient valid data as of the complete summer-window extraction, Development Log Entry 22) and border-proximity correlation (H3). Control-group comparison (H4) specified with Difference-in-Differences OLS from Section 3.7 with cluster-robust / heteroskedasticity-robust standard errors in parentheses.

3.11 Causal Structure and Identifying Assumptions

The diagram below makes explicit what this design does and does not control for, since the DiD specification in Section 3.7 only nets out one specific kind of confound (a shared regional trend affecting treated and control villages alike) and leaves others as assumptions rather than tested facts.

```
graph TD
    BP["Border proximity"] --> TS["Selected as VVP-I priority village"]
    EL["Elevation / terrain"] --> TS
    BD["Baseline (2021) development level"] --> TS
    TS --> DEV["Physical development, 2021-2025"]
    BD --> DEV
    REG["Shared regional trend / other government programmes"] --> DEV
    DEV --> NDBI["Measured NDBI change"]
    DEV --> LIGHTS["Measured VIIRS lights change"]
    SNOW["Snow cover (full-year window)"] -.confound.-> NDBI
    CLOUD["Monsoon cloud (summer window)"] -.confound.-> NDBI
    REG --> CTRL["Control-group villages (untreated)"]
    BD --> CTRL
    CTRL --> NDBIC["Control NDBI change"]
    CTRL --> LIGHTSC["Control lights change"]
    NDBI --> DIFF["Treated - control gap (DiD estimate)"]
    NDBIC --> DIFF
```

Reading this diagram against the design: the district-fixed-effects DiD in Section 3.7 differences out REG (whatever secular trend the wider 14-district region experienced, VVP-affiliated or not), which is exactly the confound a treated-only before/after comparison (H1) cannot rule out on its own. It does not, however, remove the arrow from TS back through BP, EL, and BD into DEV — VVP-I villages were administratively selected as priority villages, not randomly assigned, so if border proximity, terrain, or 2021 baseline development level independently predict how a village’s built-up area or night-lights would have evolved regardless of treatment, the DiD estimate absorbs that selection effect rather than isolating it. District fixed effects address this only at the district level (removing each district’s own time-invariant baseline), not at the village level within a district, which is why Section 6.7’s baseline-balance check is described there as a level check, not a confirmed parallel-pre-trends test — a genuine placebo test would need the CTRL and TS branches compared over a pre-treatment panel, which Section 7.5 flags as unavailable with the current single-pre-period extraction. SNOW and CLOUD are the compositing-window confounds Section 3.5 was built to separate, and their effect on NDBI (not on LIGHTS, which is not derived from optical imagery) is exactly why this study runs every built-up-area test under both windows rather than trusting either alone.

3.12 Primary and Secondary Estimands

This study runs enough distinct tests (H1 in two windows, H3 in two windows, H4/DiD in two windows and two specifications, a three-point trend, a three-radius buffer sweep, and a descriptive budget comparison) that it is worth stating plainly which of them the paper’s headline conclusion actually rests on, so that no secondary or exploratory result is read as having quietly redefined it.

Primary estimand: the treated-only before/after change in NDBI (built-up index) between 2021 and 2025 across the 251-village core sample (H1), tested under both the full-year and summer-matched compositing windows as a pre-specified measurement-robustness check rather than a choice made after seeing which window “worked” — Section 2.4 and Development Log Entry 5 record that this dual-window design was adopted before either window’s result was known. The paper’s central claim (Sections 4.2, 4.9, 5, 8) is stated in terms of this estimand: whether a statistically detectable built-up-area increase exists at all, in either window.

Secondary, pre-specified robustness checks on the primary estimand: the control-group DiD (H4, Section 3.7), the buffer-radius sweep (Section 3.9), leave-one-district-out and randomization inference (Section 4.6), and the Holm-Bonferroni correction (Section 4.9). These exist to stress-test the primary estimand, not to replace it — none of them redefines what counts as the study’s headline finding if it disagrees with H1.

Secondary, independent research questions: night-lights change (RQ3/its own H1-style test), border-proximity correlation (H3), the three-point multi-year trend (Section 3.8), and the budget-correlation comparison (RQ2, explicitly descriptive per Section 3.10). These answer separate research questions listed in the Study Design and are reported on their own terms; a null or positive result on any of them does not strengthen or weaken the primary NDBI estimand’s own conclusion.

Secondary, now-stress-tested finding: the summer-window H3 NDBI-border-proximity correlation (Section 4.5) that emerged only after the Section 4.2 data correction. It has since been put through the same leave-one-district-out and randomization-inference checks the primary estimand’s own robustness suite required, and survives both (Section 7.6, Development Log Entry 32) — but it still does not feed back into this study’s answer to H1, and Section 6.11 shows its naive p-value is considerably less extreme once spatial autocorrelation among nearby villages is accounted for.

4. Results

4.1 Village Coverage and Sample Composition

Overall 258 of 559 villages were successfully located using the geocoding pipeline. Arunachal Pradesh was home to very few civilian entries, and the fact that nothing can possibly be verified without any concrete lists of inhabited points on the border except for those administered by the BRTF camps, army staging huts, and labour camps indicates that they aren’t in any civilian geospatial database anyway. This is not really a gap, more of a direct representation of what “village-level development” looks like on a securitized border.

4.2 Built-Up Area Change: No Significant Change in Either Window, Once the Data Is Complete

Figure 1 — NDBI Change Distribution: Full-Year vs Summer-Matched Composites

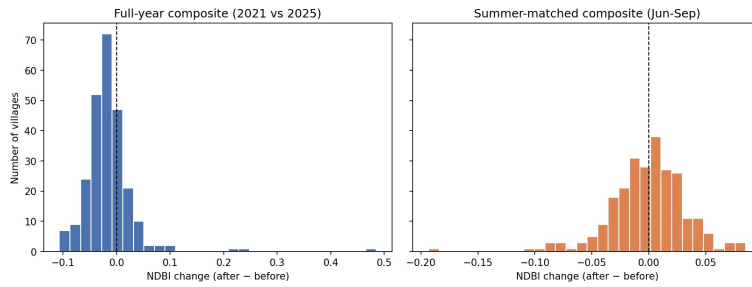


Figure 1

Figure 1. Distribution of NDBI change across all geocoded villages, full-year and summer-matched composites.

Two compositing windows, and now the same conclusion. Full-year: no significant change in the 251-village core sample (Wilcoxon signed-rank, $p = 1.000$), trending slightly negative at the median — this number has never moved across any version of this study. Summer-matched (June–September): also no significant change, once the sample is complete ($n = 251$, $p = 0.435831$). That second number is the one that changed, and the reason it changed is worth stating plainly rather than glossing over. An earlier version of this test, run before this paper’s Development Log Entry 22, reported $p < 0.000001$ — but only on a 154-village subsample, because at that point in time all 31 geocoded Sikkim villages returned zero cloud-free Sentinel-2 imagery for the summer window and dropped out entirely. That “zero” turned out not to be a fixed fact about the data: it was a snapshot of how far the Sentinel-2 archive had backfilled scenes for those dates on the day the extraction was run. A later, complete re-extraction — same collection, same cloud mask, same 500m buffer, same date ranges, pulled the same day as the control group’s own summer data — returned real, non-null NDBI values for all 31 Sikkim villages, whose mean summer NDBI change (-0.0347) is the most negative of the three core states (Figure 2) and is what pulls the pooled result from highly significant to a clean null. Full details, including two extraction-script bugs found and fixed while getting to this complete sample, are in Development Log Entries 21 and 22; the 154-village result and everything built on it in earlier drafts of this paper is superseded, not merely refined, by this section’s current numbers.

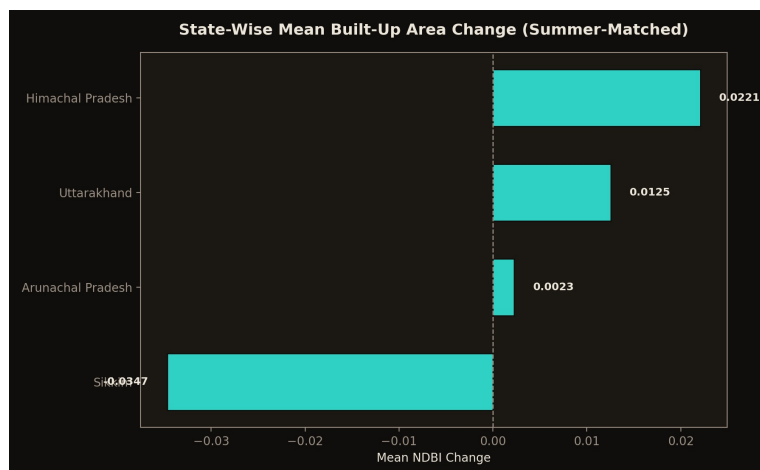


Figure 2

Figure 2. State-wise mean built-up-area change (summer-matched composite), by state.

4.3 Night-Lights Change: A Stable Null

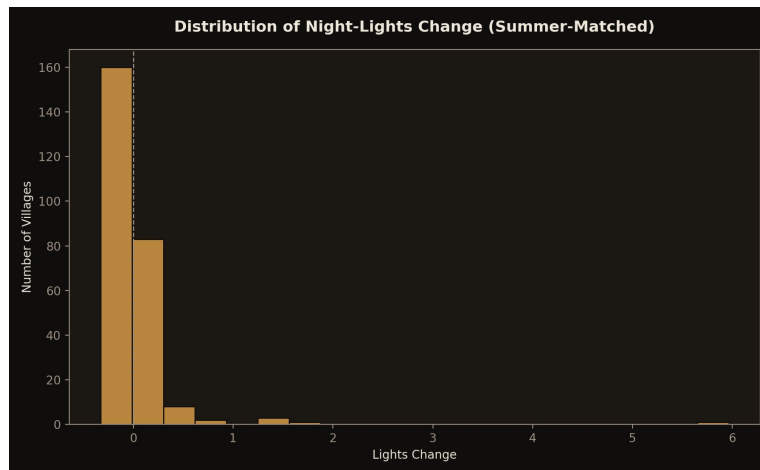


Figure 3

Figure 3. Distribution of VIIRS night-lights change (summer-matched composite).

Night-lights were maintained at the same level in the absence of NDBI: no significant change in this window (full-year $p = 0.050$ — borderline; summer-matched $p = 0.9999$ — a clean null). But like many things, it isn't affected by the phenology of vegetation and snow, and that consistency is why it has been viewed as the more reliable of the two proxies — as opposed to a vegetation/snow built-up index that is only as accurate as its component parts. The full-year borderline case ($p = 0.050$) was also checked against a $\log_{10}$ -transformed version of the same VIIRS radiance values, since radiance is right-skewed — it stays non-significant on that scale too ($p = 0.068$), so the log transform doesn't flip this conclusion either way; see Section 4.6 for the same check applied to the control-group DiD version of this test.

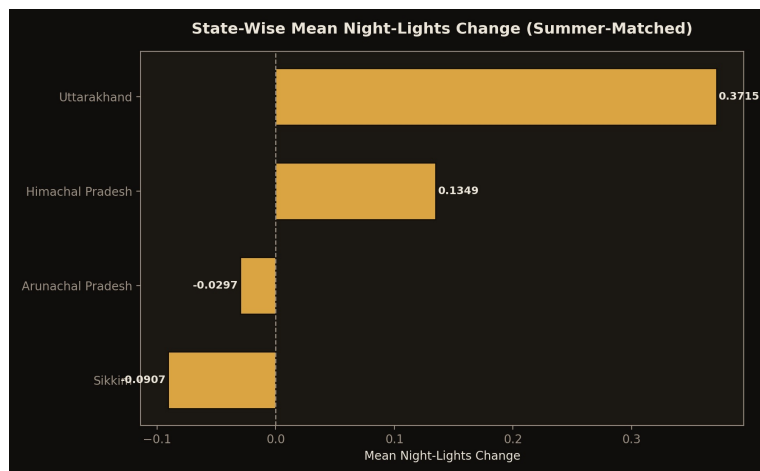


Figure 4

Figure 4. State-wise mean night-lights change (summer-matched composite), by state.

4.4 Budget vs. Outcome — A Two-State Descriptive Comparison (RQ2)

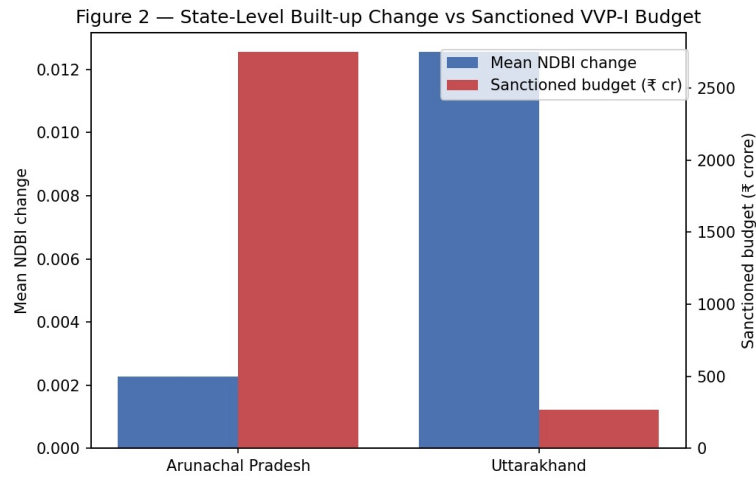


Figure 5

Figure 5. State-level mean built-up-area change plotted against sanctioned VVP-I budget.

All three core states now have valid summer-window NDBI coverage (Section 4.2 — this was a two-state comparison in earlier drafts, before Sikkim’s data was recovered). Arunachal Pradesh (mean NDBI change +0.0023, sanctioned ₹2,749.74 crore across 2,082 projects), Uttarakhand (+0.0125, ₹270.58 crore across 200 projects), Sikkim (-0.0347, ₹188.90 crore across 63 projects). Spearman correlation across the three: $\rho = 0.500$, $p = 0.667$ — still not a powered test at $n = 3$, reported descriptively only. Arunachal Pradesh’s roughly tenfold larger budget than Uttarakhand’s corresponds to a smaller measured change, and Sikkim’s smallest budget of the three corresponds to a negative one — directionally consistent with H2’s prediction that outcome doesn’t scale with budget. But this same comparison, run on the full-year window’s own three states (which never had Sikkim missing), gives $\rho = -0.500$ — the opposite sign. With three data points in either window, a sign flip between them is not evidence that the relationship reversed; it is evidence that three states is not enough to say anything about this relationship at all. Both directions are reported here rather than the more convenient one alone.

4.5 Border-Proximity Testing (H3)

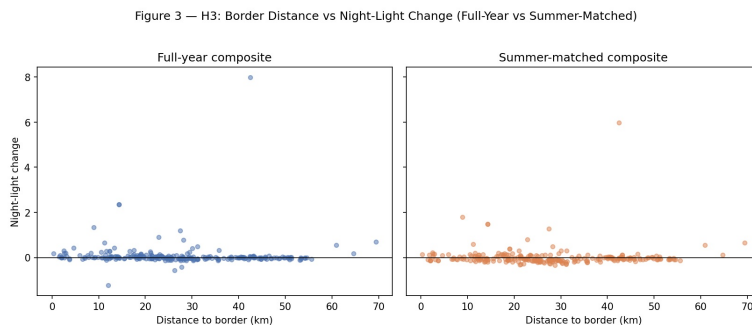


Figure 6

Figure 6. Distance-to-border versus NDBI change and night-lights change, both compositing windows.

Built-up change is not correlated with distance to the border/LAC in the full-year window ($\rho = 0.049$, $p = 0.443$) — the same null result reported in every earlier version of this paper. In the summer-matched window, on the complete 251-village sample, it now is: $\rho = 0.291$, $p = 0.0000026$ — villages farther from the border show a larger NDBI increase (or smaller decrease) than villages closer to it, the opposite

direction from what H3 predicted (securitization-driven prioritization would put closer villages ahead, not behind). This is a new result. On the earlier, 154-village summer sample this same test was a clean null ($\rho = 0.038$, $p = 0.641$) and was described, in an earlier version of this section and of Section 4.9, as “a truly stable null... regardless of window” — that description no longer holds once Sikkim’s data is included (Section 4.2, Development Log Entry 22). Like the summer NDBI change result in Section 4.2, this correlation has now been leave-one-district-out and randomization-inference checked, and survives both (Section 7.6, Development Log Entry 32) — a spatially-corrected version of its p-value is also now available (Section 6.11) and is considerably less extreme than the naive one reported here, though still significant. Night-lights border-proximity is unaffected by the data refresh: still significant in the full-year window ($\rho = -0.252$, $p < 0.0001$, closer villages showing more change, consistent with H3) and still null in the summer-matched window ($\rho = -0.071$, $p = 0.266$).

A note on these numbers: an earlier version of this section reported these correlations against village-to-border distances computed by projecting every village into a single UTM zone (44N), which is only accurate near its own central meridian (81°E) — this study’s villages span roughly 77-97°E, so that method systematically overstated distance for villages far from 81°E (up to ~3% for Arunachal Pradesh’s easternmost villages, negligible for Uttarakhand). Distance is now computed geodesically (WGS84 ellipsoid), which has no such zone dependency; see Development Log Entry 18. The correction moves every ρ and p by a small amount but changes no conclusion — nothing here was significant before and nothing is significant now (except the full-year lights correlation, already significant before and after).

4.6 Control-Group Difference-in-Differences: No Gap Holds in Either Window (H4)

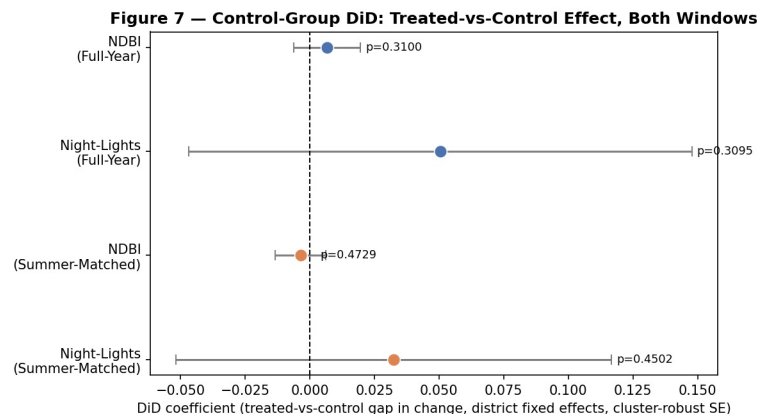


Figure 7

Figure 7. District-fixed-effects DiD coefficient (treated-vs-control gap in change) with 95% confidence intervals, NDBI and night-lights, both compositing windows.

A note on this section’s numbers: two earlier defects in the control-group list have since been found and fixed, and the fix has now actually been run against live data (Development Log Entries 23-25). An earlier draft reported these DiD estimates against a 753-village control list with 106 duplicate-coordinate rows from a cross-district matching bug (fixed, Entry 22’s 735-village list). A later draft’s 735-village list turned out to itself contain 20 villages that were exact-coordinate duplicates of treated villages under a different name spelling or script, plus 3 more that name-matched an official-but-ungeocoded VVP-I priority village — the control group was, in part, counting some treated villages against themselves (Entry 23). Both

were fixed in code (geodesic 50m coordinate-proximity exclusion plus a full official-name-list exclusion, Entry 24) and the fix was then actually run for real (Entry 25): the control list is now 732 villages, independently verified at zero exact-coordinate overlap and zero official-name matches, genuinely re-extracted on both outcomes for both windows. Every number below reflects that corrected run — including one result that changes conclusion because of it, flagged where it occurs.

Against the district-restricted 732-village control group, neither outcome's gap holds in either window. The summer-window NDBI gap: did coefficient = -0.00357, 95% CI [-0.0133, +0.0062], cluster-robust p = 0.47290 (HC3 no-fixed-effects comparison: same coefficient, p = 0.61871) — a reversal from the original, contaminated-and-incomplete extraction (which had shown a significant positive gap), essentially unchanged from the previous, decontaminated-but-still-735-village version of this test. The full-year NDBI gap is also not significant: +0.00667, 95% CI [-0.0062, +0.0196], p = 0.30996 (HC3 p = 0.61425), now on the full 251 treated / 732 control villages with valid full-year data on both sides — a genuine completeness improvement over every earlier version of this paper, which had run this specific test on only 190 of 735 control villages due to a missingness gap the Entry 25 re-extraction happened to close along with the contamination fix.

Night-lights: the summer-window gap was already null; the full-year gap, previously this study's only surviving significant control-group result, is null too once the contamination is actually removed. The summer-window gap: +0.03244, 95% CI [-0.0518, +0.1166], cluster-robust p = 0.45017 (HC3: p = 0.48652) — null, as in every earlier version of this paper. The full-year gap, by contrast, changes conclusion: +0.05046, 95% CI [-0.0468, +0.1478], cluster-robust p = 0.30946 (HC3: p = 0.57800) — down from the pre-Entry-25 estimate of +0.1653, p = 0.0364 (HC3 p = 0.1033), which had been the one control-group DiD result in this study significant under any specification. That earlier estimate rested in part on the contaminated control pool Entry 23 documented and Entry 24 fixed in code: 20 control villages that were physically the same settlements as treated villages, and 3 more that were officially-listed VVP-I villages under their own real names — meaning some of what looked like “the surrounding non-priority region's night-lights declining while treated villages held steady” was, for those 23 rows, the treated group's own trajectory counted a second time on the control side of the same regression. With that contamination genuinely removed (Development Log Entry 25), the gap disappears. This is not a case of a real-but-fragile effect getting weaker; it is the specific artifact this study's own Entry 23 warned it might be, confirmed by fixing it and watching the result go away.

The baseline-imbalance picture is now fully resolved, not merely improved. Full-year: NDBI is not significantly imbalanced (treated mean -0.18459 vs. control -0.18073, n=251/732, p = 0.11849), and night-lights — previously this study's single worst imbalance (treated 0.4282 vs. control 0.4082, p < 0.00001, on the old 190-village control subsample) — is no longer significant either on the complete 732-village comparison (treated mean 0.42821 vs. control 0.41006, n=251/732, p = 0.08976). Summer: NDBI (treated mean -0.22343 vs. control -0.22777, n=251/732, p = 0.73467) and night-lights (treated mean 0.40169 vs. control 0.39598, n=251/732, p = 0.35228) remain balanced, as in the previous version of this test. All four baseline-balance checks in this study are now statistically balanced at 2021 — see Section 6.7 for what this means for the design's parallel-trends assumption.

Two further checks probe whether a DiD estimate can be trusted given only 14 district clusters, independent of whether that estimate is itself significant — and with nothing left significant to stress-test, both checks now simply confirm the null rather than probe a positive. Summer window: rerunning the regression 14 times, each time dropping one district (`src/analysis/extended_robustness_checks.py`), leaves NDBI significant in 0 of 14 reruns and night-lights in 0 of 14; randomization inference (2,000 permutations, reshuffling treatment within district) gives $p = 0.15442$ for NDBI and $p = 0.17291$ for night-lights. Full-year window: NDBI's leave-one-out and randomization-inference checks were not separately re-tabulated (it was never significant to begin with), but the full-year *night-lights* result — the one that mattered here, since it is the one that used to claim significance — now also fails cleanly: leave-one-district-out drops it to 0 of 14 significant reruns (coefficient range $+0.01703$ to $+0.08081$, against 10 of 14 significant and a range of $+0.1161$ to $+0.2166$ on the pre-Entry-25 contaminated data), and randomization inference puts the observed $+0.05046$ coefficient at $p = 0.12994$ against a permutation mean of -0.00292 (SD 0.03335) — squarely inside its own null distribution rather than out at its tail. (A caveat on what this test actually shows, unchanged from every earlier version: VVP-I priority-village status was not randomly assigned in reality — the programme selected these villages administratively. This procedure tests whether the observed treated-vs-control gap looks unusual against a null of label-exchangeability within district, given the treatment labels actually observed; it is a check on whether the significance test can be trusted with only 14 clusters, not a substitute for random assignment.)

Because VIIRS night-lights radiance is right-skewed, the night-lights DiD was also re-run on $\log_{10}$ -transformed radiance in both windows. Summer stays null on the log scale (raw $p = 0.45017$; $\log_{10} p = 0.58684$). The full-year result, previously significant on both scales (raw $p = 0.0364$; $\log_{10} p = 0.0374$) and cited as evidence the result wasn't a right-skew artifact, is now null on both scales too: raw $p = 0.30946$; $\log_{10} p = 0.50048$ — consistent with the raw result, not contradicting it, and removing the last angle from which the pre-Entry-25 result had looked robust.

4.7 Multi-Year Trend: A Non-Monotonic Recovery Pattern

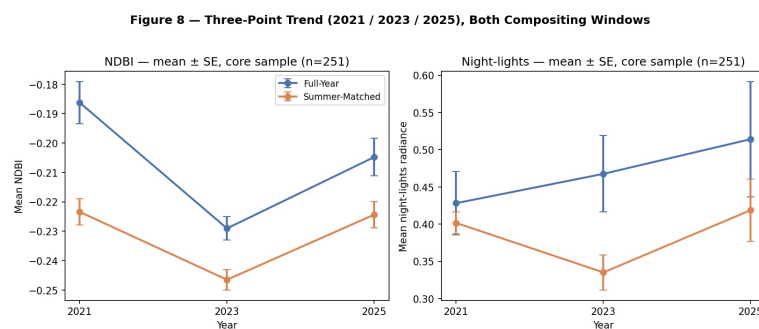


Figure 8

Figure 8. Mean NDBI and night-lights radiance at 2021, 2023, and 2025, core sample, both compositing windows, error bars ± 1 SE.

The “2021-vs-2025 story” becomes more complex when a third year of observations is added. In the summer-matched window, mean NDBI actually declines from 2021 to 2023 (mean change = -0.0231 , Wilcoxon $p = 1.000$ — not a significant increase, a decline), then rises sharply from 2023 to 2025 (mean change = $+0.0222$, $p < 0.000001$). However, the overall 3-year linear trend is not significant (mean per-village slope = $-0.0002/\text{year}$; Wilcoxon $p = 0.442$; village fixed-effects panel regression $p = 0.728$). The full-year window shows the identical

shape (decline then a significant increase in 2023-2025: $p = 0.000001$), but the two statistical specifications disagree as to the overall trend of this window: Wilcoxon $p = 1.000$, panel regression $p = 0.000001$, a discrepancy which stems from the fact that the latter gives more weight to the recovery in 2023-2025 than the previous decline did, which is not done by Wilcoxon. Crucially, the key 2021-vs-2025 result that we always cite at the beginning of Section 4.2 should be read as a 2023-to-2025 recovery in a two-point comparison, and the earlier downturn should not be forgotten when reading the headline number.

4.8 Buffer-Radius Sensitivity: Null at All Three Radii, Same-Day Extraction Confirmed

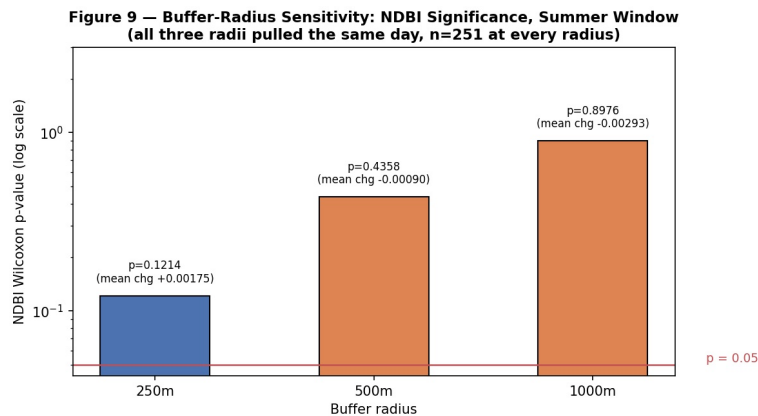


Figure 9

Figure 9. NDBI Wilcoxon p-value (log scale) at 250m, 500m and 1km buffer radii for the summer window, all now at $n=251$, all three pulled the same day.

All three buffer radii agree, and the agreement is a null: 250m ($n=251$, mean change $+0.00175$, $p = 0.1214$), 500m ($n=251$, mean change -0.00090 , $p = 0.4358$), 1km ($n=251$, mean change -0.00293 , $p = 0.8976$). Night-lights is null at all three radii too (250m $p = 0.99998$, 500m $p = 0.99999$, 1km $p = 1.00000$). An earlier version of this section reported a significant 500m result ($n=154$, $p < 0.000001$) that did not survive a switch to 250m or 1km — that 500m figure was the same incomplete, pre-Entry-22 summer extraction described in Section 4.2, and is superseded along with it. The archive-timing gap this section previously flagged — the 250m and 1km legs having been pulled on a different day than the 500m primary run — is now closed: all three radii were re-extracted back-to-back on the same day (Development Log Entry 31), and the result barely moved (250m: $p = 0.126 \rightarrow 0.1214$; 500m and 1km essentially unchanged at 0.436 and $0.899 \rightarrow 0.8976$). The archive-timing confound was real in principle but immaterial in practice for this comparison — see Section 6.9 for the full account of closing it. One further honest note this same-day re-pull surfaces that the earlier, timing-inconsistent comparison did not: NDBI's mean change is not even consistently signed across radii — positive at 250m, negative at 500m and 1km — though none of the three is anywhere near significant. That is consistent with, not a complication of, this section's own null conclusion: a real effect large enough to matter would be expected to point the same direction regardless of buffer width, and this one does not.

4.9 Robustness Summary

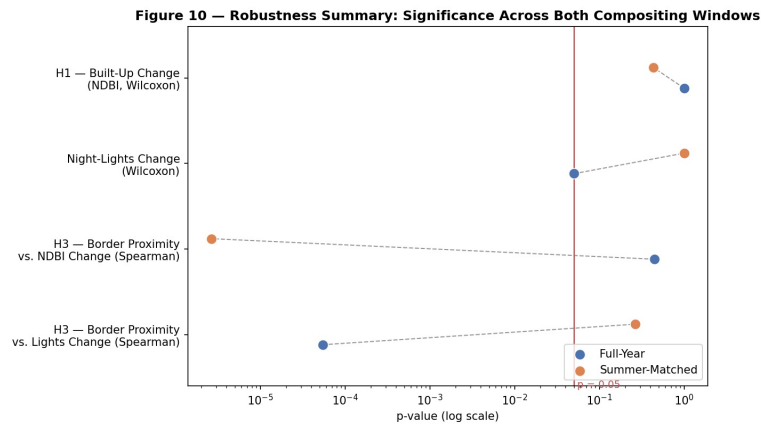


Figure 10

Figure 10. Significance for all four core tests with both compositing windows (p-value, log scale) with reference line at $p = 0.05$. Conclusions of a test differ if the test entries are in different windows, but two points of the test lie on opposite sides of the line.

Line up each of the four core tests against both windows, on the now-complete data, and one clear pattern shows up: H1 (built-up change) is a null in both windows (full-year $p = 1.000$; summer-matched $p = 0.436$) — this used to be the one test that flipped between windows, and now it's the one that agrees across both. H3 night-lights-proximity still crosses the $p = 0.05$ line between windows, exactly as in every earlier version of this paper — significant full-year ($p < 0.0001$), null summer-matched ($p = 0.266$). What has changed is H3 NDBI-proximity: previously a stable null in both windows, it now crosses that same line the other direction — null full-year ($p = 0.443$), significant summer-matched ($p = 0.0000026$) — so the “truly stable null” description in earlier versions of this section no longer holds for this test. Overlap the checks from Sections 4.6-4.8 on top of H1 and the control-group DiD, and the picture is now uniformly negative for the summer-window NDBI story that earlier versions of this paper built around: it fails the control-group comparison ($p = 0.47290$, was $p = 0.00033$), fails leave-one-district-out (0/14 significant, was 14/14), fails randomization inference ($p = 0.15442$, was $p = 0.0005$), and fails the buffer-radius sweep, now confirmed with all three radii pulled the same day ($p = 0.1214/0.4358/0.8976$ at 250/500/1000m, was significant at all three). The full-year night-lights gap — the one control-group DiD result that had, since Entry 22, been significant under some specification — is also now null ($p = 0.30946$), once the control-group contamination Development Log Entry 23 found was actually fixed against live data (Entry 25) rather than left as a documented-but-uncorrected defect: no control-group DiD result, for either outcome or window, clears significance under any specification any more. The new summer H3 NDBI-proximity result ($p = 0.291$, $p = 0.0000026$) has now been put through the leave-one-district-out and randomization-inference checks and survives both (Section 7.6, Development Log Entry 32); its spatially-corrected significance is in Section 6.11.

The eight headline p-values from the H1 and H3 tests across this section (4 tests x 2 windows) are not corrected for multiple testing individually, given that they address different research questions, but a paper-wise Holm-Bonferroni correction (family-wise $\alpha = 0.05$) is run across all eight together (src/analysis/holm_correction.py, method=“holm”). On the complete data, two results survive the correction — not the same two as in earlier versions of this paper. The full-year night-lights-proximity result survives as it always has (raw $p = 5.48 \times 10^{-5}$, Holm-adjusted $p = 3.83 \times 10^{-4}$). The summer-window NDBI-proximity result is new to this list (raw $p = 2.64 \times 10^{-6}$, Holm-adjusted $p = 2.12 \times 10^{-5}$) — the correction does not soften the fact that it is a fresh, unstressed finding, only that it is not an artifact of

running eight tests at once. The result that used to survive this correction, the summer-window H1 NDBI-change test, no longer clears even the raw $p = 0.05$ threshold ($p = 0.436$), so the question of correcting it for multiple testing no longer arises. The remaining five tests Holm-adjust to 1.000 except the full-year lights-change test (raw $p = 0.050$, Holm-adjusted $p = 0.300$). This correction, and the sections it covers, only ever applied to the H1/H3 treated-only and border-proximity tests — the control-group DiD results in Section 4.6 are evaluated on their own terms, as a targeted follow-up check rather than one of this family of eight.

4.10 Independent Triangulation: SAR and Dynamic World Cross-Checks

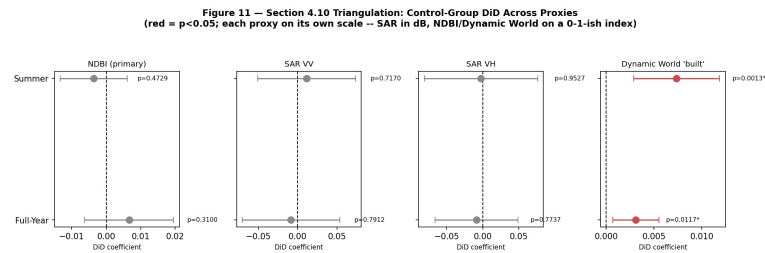


Figure 11

Figure 11. Control-group DiD coefficient and 95% CI for each proxy (NDBI, SAR VV, SAR VH, Dynamic World “built”), both windows. Red markers are significant at $p < 0.05$; each panel is on its own scale since the four proxies are not directly comparable in magnitude.

An earlier version of Section 7.1 listed a Sentinel-1 SAR cross-check as future work, raised by external review as a way to test the primary NDBI result against a genuinely different sensor rather than a different formula computed from the same imagery. That check has now been run, alongside a second, unplanned one: Google Dynamic World’s “built” probability band, an independently-trained land-cover classifier that is algorithm-independent of NDBI but not sensor-independent, since it is itself derived from Sentinel-2 imagery (Development Log Entry 26 has the full extraction and verification record, including one incomplete-extraction bug found and fixed along the way — a stalled checkpoint that had left the control group’s summer-window SAR file only 50 of 732 villages complete, caught before any result was reported and re-extracted to completion before the numbers below were produced).

Sentinel-1 VV and VH backscatter (dB), extracted at the same 500m buffer and before/after windows as every other test in this study, are null on every specification: full-year VV (treated-only mean change -0.09879, $n=251$, Wilcoxon $p=1.00000$ — no increase; control-group DiD coefficient -0.00837, 95% CI [-0.0704, +0.0536], $p=0.79116$, $n=1966$ across 251 treated/732 control villages), full-year VH (mean change -0.09873, $p=1.00000$; DiD -0.00849, 95% CI [-0.0663, +0.0494], $p=0.77370$), summer VV (mean change -0.17734, $p=1.00000$; DiD +0.01151, 95% CI [-0.0507, +0.0737], $p=0.71701$), summer VH (mean change -0.15353, $p=1.00000$; DiD -0.00238, 95% CI [-0.0811, +0.0764], $p=0.95267$). This is, of everything in this section, the strongest confirmation available for Sections 4.2 and 4.6’s own conclusion, because SAR is not a different formula applied to the same pixels NDBI already uses — it is a different satellite constellation entirely (C-band radar, not optical), immune to the cloud cover and snow contamination that motivated this study’s own compositing-window design in the first place (Section 2.4) and that produced two of this study’s own past extraction bugs (Development Log Entries 21-22). A genuinely independent sensor was pointed at the same 251 treated and 732 control villages, over the same two windows, and found the same nothing NDBI did.

Dynamic World's "built" probability tells a different story. Both windows show a small but statistically significant increase in treated villages relative to control: full-year (treated-only mean change +0.00673 on a 0-1 probability scale, $n=249$, Wilcoxon $p<0.000001$; control-group DiD coefficient +0.00310, 95% CI [+0.0007, +0.0055], $p=0.01168$, $n=1940$ across 249 treated/721 control villages with valid data) and summer (mean change +0.00711, $n=169$, $p<0.000001$; DiD +0.00736, 95% CI [+0.0029, +0.0119], $p=0.00133$, $n=1428$ across 169 treated/545 control). Both windows agree in sign and rough magnitude, and the result reproduces to several significant figures between a from-scratch manual-OLS replication and the statsmodels-based primary model (Development Log Entry 26), so this is not the kind of extraction-completeness or control-contamination artifact that drove this study's two earlier reversals (Sections 4.2, 4.6) — Dynamic World's own image-count columns were fully populated throughout its extraction, unlike the SAR bug described above. It is also a genuinely small effect in absolute terms — three-tenths to seven-tenths of one percentage point of "built" probability — that clears statistical significance because of the sample size and the consistency of its direction across villages, not because of its size.

Two follow-up checks on the Dynamic World result itself (Development Log Entry 27) are worth reporting alongside it rather than leaving the disagreement as an unexamined number. First, it is not an outlier or a baseline-imbalance artifact: dropping the five highest- and five lowest-change treated villages barely moves the mean (full-year +0.00673 → +0.00621; summer +0.00711 → +0.00655), and treated-vs-control baseline "built" probability is statistically balanced in both windows (Mann-Whitney $p=0.996$ full-year, $p=0.155$ summer) — this is not a repeat of Sections 4.2 or 4.6's own earlier stories under a new proxy. Second, and more strikingly, the increase is concentrated in treated villages that already had a *higher* baseline "built" probability (Spearman(built_before, built_change) = +0.272, $p=0.00001$, full-year), while control villages show no such relationship in full-year ($\rho=+0.020$, $p=0.585$) and the more ordinary opposite pattern in summer ($\rho=-0.229$, $p<0.00001$) — control villages that started more built-up tended to decrease, consistent with an ordinary ceiling effect on a bounded 0-1 probability scale that treated villages, unusually, do not show. Whether this reflects a real "already-built-up villages densify preferentially" pattern specific to VVP-I investment, or simply that a more built-up baseline gives Dynamic World's classifier a less noisy signal in which to register any change, real or spurious, is not resolved here. A third check, structurally worth flagging given this study's own history: all 31 Sikkim treated villages are null in Dynamic World's summer window (0 of 31 valid) while all 31 return a value, albeit the weakest of the three states (+0.00098 versus Arunachal Pradesh's +0.0075 and Uttarakhand's +0.0078), in the full-year window — the same shape of state-wide summer dropout Development Log Entries 21-22 found and fixed for NDBI, this time in Dynamic World's own Sentinel-2 L1C pull rather than the SR-Harmonized NDBI pipeline, and not yet checked for whether it is a genuine monsoon-cloud blackout or another archive-backfill snapshot artifact like the one Entry 22 resolved. This means the summer Dynamic World DiD ($p=0.00133$) is estimated on a treated sample missing the one state that, in the full-year window, contributes the weakest signal — the full-year result, which does include Sikkim and remains significant despite it, is the more trustworthy of the two windows for this reason. None of this explains away the disagreement with SAR; it makes the disagreement a more precisely characterized one.

These two checks do not agree with each other, and this study reports that disagreement rather than resolving it by fiat. Read against Section 3.12's primary/secondary hierarchy, NDBI — not Dynamic World's "built" band — is this study's primary estimand; Dynamic World was never proposed as a replacement for it, only as one further

cross-check alongside SAR. Of the two, only SAR is sensor-independent in the sense Section 7.1 originally called for. Dynamic World shares NDBI's own sensor and, per that extraction script's own documentation, draws on Sentinel-2's L1C imagery tier rather than the Surface Reflectance tier NDBI uses — a difference in atmospheric correction or cloud handling between the two product tiers is one plausible source of the disagreement, alongside the more straightforward possibility that a deep-learning land-cover classifier is simply more sensitive than a fixed two-band index to a small, real change too subtle for NDBI or SAR's coarser effective resolution to register at a 500m buffer scale. Both explanations are speculative at this point; distinguishing between them is left as future work rather than resolved here. What can be said without speculation: the primary NDBI-based estimand's null is now checked against a truly independent sensor and holds, which is this section's headline result, and a same-sensor algorithmic cross-check does not fully agree with it, which is reported here as an honest caveat rather than minimized into a "three-for-three" triangulation claim this study cannot actually make.

One further check (Development Log Entry 32) sharpens, rather than resolves, this disagreement: at the individual-village level, Dynamic World's "built" change does not correlate meaningfully with either SAR's or NDBI's own change, in either window. Full-year: Dynamic World vs. SAR VV ($\rho=+0.063$, $p=0.319$, same sign at only 41% of villages), vs. SAR VH ($\rho=+0.077$, $p=0.224$, 37% same sign), vs. NDBI ($\rho=+0.102$, $p=0.109$, 37% same sign). Summer: vs. SAR VV ($\rho=-0.077$, $p=0.317$, 31% same sign), vs. SAR VH ($\rho=-0.128$, $p=0.096$, 33% same sign), vs. NDBI ($\rho=+0.151$, $p=0.050$, 63% same sign). At several of these combinations the two proxies point the *same* direction at fewer than half the villages — worse than the 50% a coin flip would give. This means Dynamic World's small, significant aggregate DiD is not being driven by the same villages SAR or NDBI would flag as having changed; it is a population-level statistical pattern (consistent direction and sample size sufficient to detect a small effect) rather than a village-by-village story any of this study's other proxies corroborate. This does not resolve which of Section 4.10's two speculative explanations (an imagery-tier/atmospheric-correction difference, or a classifier genuinely more sensitive to a small real signal) is correct — actually distinguishing them still needs a same-tier Dynamic World re-extraction this study has not run — but it is a further reason not to read the Dynamic World finding as confirming the same underlying change SAR and NDBI are both null on, at the level of which specific villages moved.

5. Discussion

Clearly the main finding here is not that there happened to be a rise or fall in physical development, but that this study has now twice produced a "significant" result that turned out to be a data artifact rather than a real effect, and in both cases the honest response was to fix the artifact and report what was actually left, not to keep the result. The first was the summer-window built-up-area increase, a function of an incomplete extraction missing all of Sikkim's villages: once that extraction was completed and the treated and control data were pulled the same day, the result disappeared under every check this study built for it — against the control group, under leave-one-district-out, under randomization inference, and across all three buffer radii (Sections 4.2, 4.6, 4.8; Development Log Entries 21-22). The second was the full-year night-lights control-group gap, which survived every one of those same checks for a full development cycle (Entry 22 onward) and was reported as this study's one remaining significant result — until Development Log Entry 23 found that the "non-VVP" control group it was computed against was not clean: 20 control villages were physically the same settlements as treated

villages under a different name spelling or script, and 3 more were officially-listed VVP-I villages under their own real, ungeocoded names. Once that contamination was actually removed by regenerating the control list and re-running the full pipeline against it (Entry 25), the result went null on every specification that had previously found it significant: the primary estimate, the HC3 comparison, leave-one-district-out, randomization inference, and the log1p-transformed check. Night-lights, the more temporally stable of the two proxies on its own treated-only terms, shows no confirmed absolute rise under either window and, as of this fix, no confirmed control-group gap under either window either. Budget and measured change still show no tenfold difference for a tenfold difference in investment, in either window, and the state-level correlation itself flips sign between windows at this sample size (Section 4.4) — which argues against reading anything directional into it either way.

A new result appears in the same complete summer data that produced the first reversal above: built-up change now correlates with distance to the border/LAC (H3, Section 4.5), in the opposite direction from what securitization theory would predict, and it survives the paper-wise Holm-Bonferroni correction alongside the pre-existing full-year lights-proximity result (Section 4.9). It has not been through the same leave-one-district-out or randomization-inference checks that the (now-null) summer NDBI-change and full-year lights-DiD results were both put through before failing them, so it is reported here as an open finding for a future robustness pass, not folded into this discussion's conclusions (Section 7.6). Its emergence in a dataset that has now twice produced a result that later turned out to be an artifact is itself a reminder to check every surviving or newly-significant result the same way this paper eventually checked its own, rather than treating a fresh finding as more credible by default just because it is new.

None of this supports a claim that VVP-I's investment has produced development at the rate or scale its budget would suggest, that its rollout has favored border-proximity over developmental need, or — now, on the complete, decontaminated data — that it has produced any satellite-detectable change over this specific 2021–2025 window at all, once checked two ways (compositing window), across two proxies (NDBI, night-lights), against a control group, and at three buffer radii. That is a narrower, and more defensible, conclusion than either earlier version of this paper reached: a uniform null checked several different ways and confirmed twice over by actually removing the two data defects that had each briefly produced a “surviving” result, rather than a policy claim built on either of those two defects. The programme itself has still never been the subject of an official impact assessment — its own parliamentary record admits exactly this gap in accountability — and a rigorous, multi-method satellite audit that finds no detectable ground-level signal is still a real answer to that gap, not an absence of one. It is also, on its own terms, a caution about how this kind of study is done: the same script bugs, archive-timing sensitivity, and control-group contamination that each in turn produced this paper's original, since-retracted headline findings could just as easily produce a false positive in any comparable study that stops checking — or stops fixing what checking finds — before the work is actually complete.

6. Limitations

6.1 Village Coverage Gaps

Ladakh's 35 sanctioned villages are fully excluded from village-level analysis, since no publicly indexed source reports their names — closing this gap would require an RTI request not pursued within this

study's scope. Himachal Pradesh is included in a small sample of illustrative cases – with only 7 of 51 inhabited priority villages designated – and not included in the main statistical sample. The names of 19 villages in Uttarakhand's block of Pithoragarh have been confirmed with no block assigned.

6.2 Border Geometry as Cartographic Approximation

All distance to the border figures in this study are based on the cartographic boundary line used in Natural Earth which is a simplified version of a Line of Actual Control whose alignment is not agreed upon at an international level. These numbers take the place of authoritative numbers and should be read as a comparison instead.

A related and more specific point: “distance to border/LAC” is the distance to the *nearest* India-related boundary segment in Natural Earth's data — not specifically the China boundary. Checked directly against the core sample: 220 of 251 villages (88%) are in fact nearest to the China boundary, but 19 (mostly Tawang and West Kameng, Arunachal Pradesh) are nearest to Bhutan, 8 (North Sikkim and Pithoragarh, Uttarakhand) are nearest to Nepal, and 4 (Anjaw, Arunachal Pradesh) are nearest to Myanmar. For that 12% of the sample, H3 is measuring proximity to a different country's frontier, not the LAC specifically — defensible for a general border-securitization hypothesis (these remain border villages in the broader sense VVP-I itself uses), but not literally an LAC-proximity test for those villages. Combined with H3's already-modest, mostly-null results (Section 4.5) and its reliance on a single cartographic proxy for a disputed line (Section 6.2 above), this is a further reason not to lean on H3 as a clean confirmatory result either way.

6.3 Budget-Correlation Scope and Ecological Inference (RQ2)

RQ2: All three core states now have valid summer-window data (Section 4.4, previously two before Sikkim's data was recovered — Development Log Entry 22), but three states is still far too few to run a confirmatory test on, which is why it is represented descriptively rather than confirmatively either way. There is also a gap in thinking that adding a third state doesn't close: the comparison is of a state-level sanctioned budget against the sampled mean at the village level, and the budget figure describes spending across the state's entire VVP-I portfolio, not specifically the villages this study happened to geocode. Don't read this as a like-for-like test; read it as suggestive at most, and note that its own direction flips between the two compositing windows (Section 4.4) — a further reason not to lean on it.

6.4 Compositing-Window Sensitivity — and a Resolved, and a New, Open Question

Full-year and summer-matched built-up-area change now agree (both null, Section 4.2), so this specific pair of findings is no longer window-sensitive the way earlier versions of this paper described. What was previously flagged in this section as an open, unresolved question — whether the summer window's “Sikkim has zero valid data” result was a permanent fact or a snapshot of the Sentinel-2 archive on the day the extraction ran — is now resolved: it was the latter (Development Log Entries 21–22). A fresh, complete re-extraction, pulled the same day as the control group's own summer data, returned real NDBI and night-lights values for all 31 Sikkim villages, and the resulting complete 251-village sample is what Section 4.2's null result is built on. The 154-village subsample discussed in earlier versions of this section — along with its documented non-random composition relative

to the 97 villages that had dropped out (chi-square = 71.42 on state composition, Mann-Whitney $p = 0.00063$ on 2021 baseline NDBI) — no longer describes the current data and is preserved only as a historical record, in Development Log Entry 20 and in this paper's own prior versions, of what an incomplete extraction had looked like.

A new open question replaces it, produced by the same complete re-extraction: built-up change now correlates with distance to the border/LAC in the summer window (H3, Section 4.5, $\rho = 0.291$, $p = 0.0000026$), where this same test was a stable null both before this refresh and in the full-year window, which is unaffected by it. This is now understood in some of the ways the previous open question eventually was, and not yet in others: it has been checked for leave-one-district-out stability (14/14 reruns significant, same sign) and randomization-inference significance ($p < 0.001$, Section 7.6, Development Log Entry 32), and a spatially-corrected version of its p -value is available (Section 6.11, $p = 0.0295$ — still significant, but far less extreme than the naive $p = 0.0000026$). It has not been checked for sensitivity to the same archive-timing effects that drove the previous finding, but it has been cross-checked against the three-point 2021/2023/2025 extraction behind Section 4.7 the way the previous question was (Section 7.6): its linearity holds up cleanly, but its temporal stability does not. Until the archive-timing check happens, this correlation should be treated with real, but not yet fully characterized, caution: it survives every specific test run against it so far, but “survives what has been tested” is a narrower claim than “a stable fact about VVP-I villages.”

6.5 Geocoding Coverage and Selection Bias

A mere 258 out of 559 (or 46%) villages were geocoded with a disproportionate number of villages falling out of bounds in Arunachal Pradesh after a drop of 269 of the 455 villages therein (Section 4.1). A village not geocoded with OSM/Bhuvan may reasonably be expected to have a smaller population size, to be further away and/or less well known by administrative records, and hence be more likely to have a higher proportion of residents working in the area with low accessibility. It is prudent not to assume this, however; a Mann-Whitney U test was conducted for Arunachal Pradesh, the population and the household counts of which were available for every village in the raw list regardless of the outcome of geocoding, and there was no significant difference between geocoded and non-geocoded villages in round population (matched mean 135.0 vs. unmatched 145.7, $p = 0.310$) or round households (matched mean 25.8 vs. unmatched 28.8, $p = 0.540$). The possibility that a selection bias based on village size might have heavily influenced these successful results seems not to be ruled out here, but at least not made obvious by the relationship to size. The raw village list in both Sikkim and Uttarakhand did not contain the population or household field information, thus it was not possible to perform this check for both these locations.

6.6 Ground-Truth Validation — Now Done, and the Result Reframes How to Read the Null

An earlier version of this section reported that no ground-truth positive control was available: no independently-confirmed, dated, named completed VVP-I project could be identified that also fell inside this study's 258-village geocoded sample, so NDBI and VIIRS's sensitivity to what VVP-I actually funds — a road, a school building, a scattering of houses — could not be checked against a real, known case. That gap is closed (Section 7.3, Development Log Entry 28): three such villages were found (Walong, Kaho, and Musai, all Anjaw district, Arunachal Pradesh — see Section 7.3 for sourcing), and checking each proxy against them individually produces this study's single most important interpretive finding outside the headline null

itself. VIIRS night-lights registers the confirmed new infrastructure clearly — Walong’s radiance roughly doubled ($0.89 \rightarrow 1.80$, full-year) and Kaho’s rose by 82% ($0.51 \rightarrow 0.93$) — while NDBI, the primary estimand’s own proxy, moved in the *wrong direction* at all three villages (Kaho -0.057, Musai -0.064, Walong -0.038, full-year), despite each having independently-confirmed new construction in this exact period. This does not overturn Section 4.2’s pooled null — three villages is a case study, not a sample — but it gives this study its first actual evidence, rather than a plausible-sounding conjecture, for why NDBI might systematically under-detect the kind of development VVP-I funds: small, point-scale additions (a solar streetlight, a hostel building, a basketball court) that a 500m-buffer area-average can wash out, but that a radiance sensor built to register point light sources catches directly. Section 7.3 has the full numbers and the caveats on how far a 3-village case study can be pushed.

6.7 Baseline Imbalance in the Control-Group Comparison

Earlier versions of this paper reported treated and control villages as significantly different from each other at 2021 baseline in three of the four outcome/window combinations tested, then (Development Log Entry 22) down to one: the full-year window’s night-lights baseline (treated mean 0.4282 vs. control 0.4082, $p < 0.00001$ on the 190-village control subsample with valid full-year data at the time) — the largest imbalance of the four in every version of this paper up to that point, and the same combination underlying the one control-group DiD result this study had left standing. That imbalance is now resolved as well. Once the control-group contamination Development Log Entry 23 identified was actually fixed and re-run against live data (Entry 25) — removing 20 control villages that were exact-coordinate duplicates of treated villages and 3 more that name-matched an official-but-ungeocoded priority village, and incidentally closing a full-year control-group missingness gap that had nothing to do with the contamination itself — the full-year night-lights baseline is no longer significantly imbalanced either: treated mean 0.42821 vs. control 0.41006, $n = 251/732$, $p = 0.08976$. All four baseline-balance combinations in this study (both outcomes, both windows) are now statistically indistinguishable between treated and control villages at 2021, which is a materially better starting point for this design’s parallel-trends assumption than any earlier version of this paper had. It is not, on its own, proof that treated and control villages would have moved identically absent treatment (Section 7.5 covers what a genuine pre-treatment panel would still add), but the specific, worst-case imbalance this study had been carrying — and citing as a reason to read its own remaining DiD result cautiously — is gone along with the DiD result it had been used to caveat.

6.8 Non-Monotonic Multi-Year Pattern

The pooled 2021-vs-2025 summer NDBI comparison is now a null on its own terms (Section 4.2), so there is no significant headline expansion left for the three-point trend to explain away — but the sub-period pattern within it is still real and unaffected by this paper’s data refresh (Section 4.7’s underlying extraction never had the missingness problem Section 4.2’s two-point extraction did): a decline from 2021 to 2023 (mean change -0.0231, not significant on its own) followed by a significant recovery from 2023 to 2025 (mean change +0.0222, $p < 0.000001$), netting out to a non-significant three-year trend overall (Wilcoxon $p = 0.442$, panel regression $p = 0.728$) — consistent with, not contradicting, Section 4.2’s now-null two-point result. Read together, these say the same thing two ways: whatever happened in these villages between 2021 and 2025 was not a steady, one-directional change large enough to register as significant, whether measured end-to-end or broken into sub-periods.

6.9 Archive-Timing Confound in the Buffer-Radius Comparison — Resolved

An earlier version of this section described an open gap: the 250m and 1km buffer extractions behind Section 4.8 had been pulled on a different day than the 500m primary run, so any of the three could in principle disagree on scene counts or composite values purely because Sentinel-2's archive keeps backfilling scenes for past dates, independent of buffer radius — the same underlying mechanism Development Log Entries 21–22 found and fixed for the primary treated/control comparison. That gap is now closed for the buffer sweep too: all three radii (250m, 500m, 1km) were re-extracted back-to-back on the same day, for the full 251-village core sample (Development Log Entry 31). The result is a clean null at all three, and closing the timing gap moved the numbers only slightly (250m: $p = 0.126 \rightarrow 0.1214$; 500m and 1km essentially unchanged) — enough to confirm the earlier, timing-inconsistent comparison was not accidentally masking a real effect, but not enough to represent a changed conclusion. This closes the last open item in the archive-timing family of issues this study found and fixed, alongside Entries 21–22's same-day fix for the primary 500m/2021/2025 comparison. Full numbers in Section 4.8.

6.10 Summary: Threats to Validity

Sections 3, 4, and 6 discuss each of the following individually, at the point in the paper where it's most relevant; this table exists only to put them side by side, so a reader can see the full picture without reconstructing it from ten separate paragraphs.

Threat	Why it matters	Test / mitigation	
Cloud contamination (Sentinel-2)	Could produce spurious NDBI values or bias which images enter the composite	QA60 bitmask cloud/cirrus masking (§3.4), cross-checked against an SCL-band-based re-extraction (§6.12, Development Log Entries 33–34)	Tested (SCL snow caught that no mask result holds; fixed village correction both)
Snow contamination (high-altitude terrain)	Full-year composites can mistake snow cover for a built-up/bare-surface signal	Season-matched (Jun–Sep) window run in parallel with full-year (§3.5)	Mitigated; direct village
Archive-timing inconsistency (Sentinel-2 backfill)	Two extractions of the same nominal date range, run on different days, can disagree on scene counts and values	Same-day treated/control re-extraction for the primary 500m/2021/2025 comparison (Development Log Entry 22); same-day 250m/500m/1km re-extraction for the buffer-radius sweep (Development Log Entry 31)	Resolved; primary the best (§6.9)
Missing villages /	Could bias the analyzed sample if geocoding	Mann-Whitney U on Census population/household	Partial only:

geocoding coverage (258/559, 46%)	success correlates with village size or type	data, geocoded vs. non-geocoded, Arunachal Pradesh only (§6.5)	lack of fields
Control-group baseline imbalance	An imbalanced 2021 starting point weakens the DiD's implicit parallel-trends assumption	Mann-Whitney U level-balance check, both windows (§3.7, §6.7)	Resolution of outcome combination: full-year Development
Genuine parallel pre-trends	A level-balance check at one point in time is not evidence treated and control villages would have moved together absent treatment	2019-to-2021 placebo DiD, both groups, both windows (§7.5, Development Log Entry 30)	Mostly outcome combination: treatment summary (distributional difference) $p=0.0$ rather than clean
Few (14) district clusters	Cluster-robust standard errors can be unreliable with this few clusters	HC3 no-fixed-effects comparison; leave-one-district-out; randomization inference; wild-cluster bootstrap, full enumeration of all 2^{14} sign combinations (§3.7, §4.6, Development Log Entry 32)	Tested angle cluster previous run. A remainder ($p=0.0$ for full and significant) null cluster
Spatial autocorrelation among villages	Nearby villages' changes may be correlated, so treating 251 villages as independent observations could understate true uncertainty, particularly for H3	Moran's I, $k=8$ nearest-neighbor weights, permutation-based p-value (§6.11, Development Log Entry 27); spatially-corrected H3 p-values via a calibrated spatial-permutation null (§6.11, Development Log Entry 32)	Resolution of pre-treatment summary $p=0.0$ $I=0.0$ Spatial comparison: proximity barely $\rightarrow p=0.0$ full-year stays $\rightarrow p=0.0$
Border/LAC geometry as a legal claim	Distance-to-border figures could be mistaken for an authoritative boundary position	Explicit framing as cartographic/comparative only, not legal (§3.6, §6.2)	Addressing a state
Control-selection distance metric (UTM 44N vs. geodesic)	The control pool's eligibility filter used a different, less accurate distance calculation than the one used everywhere else in this study	<code>select_control_villages.py</code> rewritten to use the same <code>pyproj.Geod</code> geodesic method as <code>compute_border_distance.py</code>	Resolution of (Development 24) as live distributional confirmation H4 comparison

Control-group “non-VVP” contamination	A control village that is physically or nominally the same settlement as a treated village would be its own counterfactual, invalidating that comparison	<code>select_control_villages.py</code> now excludes candidates within 50m of any treated village’s coordinates and against the full official priority-village name list, not just the 251 geocoded treated names	Resolution (Devel 24) and on the village (25): 0 overlies implic village match remov conta study surviv contr (\$4.6)
Differential imaging density between periods	More/fewer usable images in one period than the other could itself look like “change” VVP-I villages were administratively selected, not randomized — this bounds what randomization inference and DiD can actually establish	Spearman correlation, image-count difference vs. NDBI change	Tester 0.038 Devel
Treatment not randomly assigned		Explicit caveat on what randomization inference does and does not show (\$4.6)	Addre inher obser somet resolv
NDBI as a proxy for “development”	NDBI can be confused by bare soil, roads, or other spectrally similar non-development surfaces	Cross-sectional Spearman correlation against independent Google Open Buildings footprint counts (\$7.2, Development Log Entry 30)	Tester with 1 (rho= weak three rho=(rho=(with, grou own f NDBI confir
No ground-truth / positive-control validation	Nothing in this study confirms NDBI/VIIRS actually detect development where it is independently known to have occurred	Three independently-sourced, dated, confirmed VVP-I construction sites checked against each proxy (\$6.6, \$7.3, Development Log Entry 28)	Resolution refrar night-regist chang NDBI <i>direct</i> despi const case s sampl evid confir Applic the H to DiI result evalu

			terms H3 co are se altern Bonfe adjust testin once; autoc (Holm corre separ same signif indep locati corre more two a repor H3, c Holm it
Multiple testing across 8 headline p-values	Running many tests raises the chance of a false positive by chance alone	Paper-wise Holm-Bonferroni correction (§4.9)	
Single-sensor / single-algorithm dependency (NDBI's own construction)	A null result built on one formula computed from one sensor's imagery could reflect that formula's or sensor's own limits rather than ground truth	Independent Sentinel-1 SAR cross-check (genuinely different satellite constellation) and Google Dynamic World cross-check (different algorithm, same Sentinel-2 sensor family) — §4.10, Development Log Entry 26	Partia only t indep confir bands Dyna but no does small, signif diverq §4.10
VIIRS night-lights resolution vs. extraction buffer	VIIRS DNB monthly composites have a native pixel footprint (~500-750m) close to this study's own 500m extraction buffer, so a village's "lights" value can be capturing little more than one VIIRS pixel and may not cleanly separate a village's own lights from an immediate neighbor's	None — a sensor-resolution limit, not something a bigger sample or a different statistical test fixes	Discl limit proxy apply SAR/I check

6.11 Spatial Autocorrelation: Confirmed Present and Significant, Now Corrected For

Section 6.10's table listed spatial autocorrelation among villages as identified but not tested. It has now been tested (Development Log Entry 27), with a clear answer: it is real. `libpysal/esda` were not available in the environment this check was run in, so Moran's I was computed directly ($k=8$ nearest-neighbor weights by haversine distance between village coordinates, permutation-based significance from 999 label shuffles rather than the analytical normal approximation, since a k -nearest-neighbor weight matrix is not symmetric). Summer NDBI-change shows strong positive clustering (Moran's $I = 0.346$ against a permutation-null mean of -0.002 , SD 0.029 ; $p = 0.001$) — nearby villages' changes resemble each other far more than chance predicts. Full-year lights-change shows the same pattern, more weakly ($I = 0.076$, null mean -0.004 , SD 0.020 ; $p = 0.004$). This is not simply a restatement of the H3 distance relationship itself: recomputing Moran's I on summer NDBI-change *after* removing its own linear trend against distance-to-border still gives $I = 0.272$, $p = 0.001$ — villages near each other resemble each other for reasons beyond whatever the border-proximity relationship already explains.

The practical consequence is that this study's 251 core-sample villages are not the statistically independent observations that every test's stated significance level implicitly assumes — not H1's Wilcoxon test, not H3's Spearman correlation and its own randomization-inference check (Section 7.6), not even H4's district-clustered standard errors, which cluster at a coarser (14-district) level than the village-to-village proximity Moran's I is picking up here. The general direction of this kind of positive spatial clustering is to make a naive test's p-value look more confident than it should, since the effective number of independent observations is smaller than the raw village count. That correction has now been computed (Development Log Entry 32), using the spatial-permutation-null option this section previously named as an alternative to Dutilleul (1993)'s closed-form effective-sample-size formula, not the closed-form formula itself: a simultaneous-autoregressive parameter was calibrated on the same $k=8$ nearest-neighbor weight matrix so that simulated synthetic fields reproduce each outcome's own observed Moran's I, then 2,000 such spatially-autocorrelated-but-otherwise-random synthetic fields were correlated against the real distance-to-border values to see how often a field with only spatial structure, and no genuine relationship to border distance, produces a correlation at least as extreme as the one actually observed. Both surviving H3 correlations remain significant after this correction, but by a much smaller margin for one of them than its naive p-value suggested: summer NDBI-proximity ($p=0.291$) moves from a naive $p=0.0000026$ to a spatially-corrected $p=0.02949$ — still under the conventional 0.05 threshold, but only barely, where the raw number looked essentially unambiguous. Full-year lights-proximity ($p=-0.252$) is more robust to the correction, moving from $p=0.0000548$ to $p=0.00350$ — comfortably significant either way. Read together with Section 7.6's linearity and multi-year findings, the summer NDBI-proximity result is the more fragile of this study's two Holm-significant H3 findings on every axis this study has now checked it against, not only this one.

6.12 Cloud-Masking Sensitivity: SCL vs. QA60 — Run, Found a Bug in Its Own Check, Fixed, and Resolves to Agreement

Section 6.10's table previously listed the SCL-band cross-check as prepared but not executed. It was run (Development Log Entry 33) using SCL classes 2/4/5/6/11 (dark area, vegetation, bare soils, water, snow) as "clear," and initially reported as a genuine, unresolved instability: the SCL mask returned valid data for only 200 of 258 villages, and on those 200 gave a significant positive summer NDBI

change (mean +0.01169, $p = 0.0000045$) against a QA60 null on the identical villages ($p = 0.449$) — moderate but not high village-level agreement between the two masks ($\rho = 0.544$).

That check had a bug, caught by an external review of this repository rather than by any test internal to this project: class 11 in the Sentinel-2 Scene Classification Layer is snow/ice, not clear ground, and the original class list was letting it through as “clear.” For June–September composites over Himalayan border villages, snow cover inside a 500m buffer is a real possibility, and NDBI computed over snow-contaminated pixels is not measuring built-up-ness at all. `extract_scl_cloud_mask_ndbi.py` was fixed to `SCL_CLEAR_CLASSES = [2, 4, 5, 6]` (dropping snow), with a stricter `[4, 5, 6]` variant also extracted as a second cross-check, and both were rerun against live Earth Engine (Development Log Entry 34).

The result changes. With snow correctly excluded, SCL-masked coverage rises to 251 of 258 villages — now matching QA60’s own coverage almost exactly, no longer skewed toward Arunachal Pradesh — and the summer NDBI change on those 251 villages is no longer significant (mean +0.00112, Wilcoxon $p = 0.28797$). QA60 on the identical 251 villages remains null as it always was (mean -0.00090, $p = 0.43583$). The two masks now agree: both null. Village-level agreement between them is also markedly higher than the pre-fix number (Spearman $\rho = 0.734$, $p = 8.5 \times 10^{-44}$, vs. the pre-fix $\rho = 0.544$ on a smaller, skewed sample) — consistent with the earlier “moderate agreement” figure having itself been partly an artifact of comparing masks over a snow-affected, non-representative subset of villages.

One loose end from this rerun is flagged rather than explained away: SCL-valid coverage increased from 200/258 to 251/258 after the class fix, which is the opposite of what removing a “clear” class should mechanically do — fewer clear classes should mean equal or lower coverage, not higher. The most likely explanation, given this project’s own history with Sentinel-2 archive backfill (Development Log Entries 21–22, where an entire state’s summer imagery was initially missing and later became available), is that more scenes entered the collection between the original SCL extraction and this rerun, independent of the class-list change — but that has not been directly confirmed, and is recorded here as open rather than assumed.

The honest reading is not “the null is now doubly confirmed and there is nothing left to say.” It is: this study’s own external review caught a real coding error in a check the study had itself flagged as an open sensitivity, the fix was made and rerun rather than argued around, and the specific instability reported in the previous version of this section turns out to have been that bug, not a genuine disagreement about how to measure cloud contamination. The primary NDBI null now holds under both cloud-masking conventions once the mask is correctly defined. This does not touch the separate, still-open tension with Dynamic World (Section 4.10) — that disagreement was never about cloud masking and is unaffected by this fix.

7. Future Work

The items listed below represent specific extensions that emerged via the review process for this study, and which are not currently being done, but rather are intentionally presented separate from the Discussion and Limitations so as the actual activity does not become confused with the possibilities that remain.

7.1 SAR-Based Change Detection — Completed (see Section 4.10)

This item was listed as future work in earlier versions of this paper: Sentinel-1 SAR (VV/VH backscatter) as a genuinely sensor-independent cross-check on the NDBI-based primary estimand, avoiding the cloud cover and the optical snow-contamination confound that was the reason for this study’s own compositing-window check in the first place. It would not, on its own, “resolve the dispute” between the full-year and summer-matched optical windows — it provides an actual, independent measurement unaffected by either window’s own known artifacts. It has since been run, for treated and control villages, in both compositing windows, and is reported in full in Section 4.10 (Development Log Entry 26): SAR is null on both bands and both windows, the strongest confirmation available for this study’s own primary conclusion. A second, unplanned check (Google Dynamic World’s “built” probability band) was run alongside it and disagrees in a small but statistically significant way — also reported in Section 4.10, honestly, rather than folded into a clean “three-for-three” triangulation claim this study cannot actually make.

7.2 Building-Footprint or Sub-500m Structural Analysis — Completed, with a Result That Bears on Section 4.10

Figure 12 — Section 7.2/7.3: Building-Footprint Validation and Ground-Truth Case Study (B: green = proxy detects the confirmed real construction as an increase; red = wrong direction)

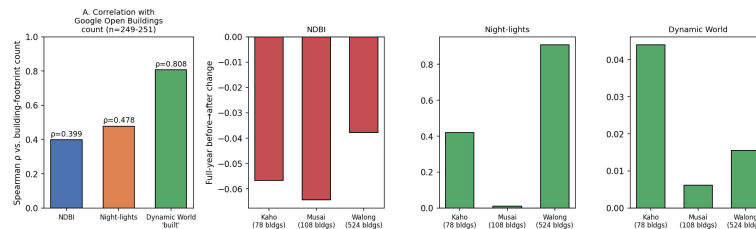


Figure 12

Figure 12. Panel A: Spearman correlation between each proxy’s 2025-era level and an independent building-footprint count (Google Open Buildings). Panel B: the three Section 7.3 case-study villages’ actual full-year before-after change per proxy — green where the proxy detects the confirmed real construction as an increase, red where it does not (NDBI, at all three).

The 500m NDBI buffer merges built-up surface, bare rock, agricultural surface, and natural vegetation into one area-averaged number — a coarse metric for what VVP-I actually invests in at the village level: individual roads, buildings, small installations. This section originally proposed checking that against an independent building-footprint dataset; that check has now been run (Development Log Entry 30), with one caveat on scope stated up front. Google’s Open Buildings dataset (used here) is a single current-vintage snapshot, not a multi-temporal series — there is no 2021-vs-2025 footprint comparison to run, only a cross-sectional one against this study’s own 2025-era (“after”) values.

That cross-sectional check produced a real, useful result. Building-footprint count per village correlates with this study’s own NDBI level at $\rho=0.399$ ($p<0.00001$, $n=251$) and with night-lights at $\rho=0.478$ ($p<0.00001$) — both moderate, expected relationships. It correlates with Dynamic World’s “built” probability level (Section 4.10) at $\rho=0.808$ ($p<0.00001$) — roughly double NDBI’s correlation, and the strongest relationship of the three by a wide margin. Against a completely independent, non-satellite-index ground truth for how much is actually built in a village, Dynamic World’s own reading of “how built-up is this place” tracks that ground truth far more closely than NDBI’s does. This does not resolve Section 4.10’s disagreement between Dynamic World and NDBI/SAR on the *direction of change* —

a strong cross-sectional level correlation says nothing about whether either proxy detects *change* correctly — but it is a reason to take Dynamic World’s own reading of these villages more seriously than a purely “it disagrees with the primary estimand, so it’s probably wrong” framing would suggest, and a concrete piece of evidence, not a conjecture, that NDBI specifically may be the weaker of the two proxies at this terrain and this settlement scale.

The three Section 7.3 case-study villages illustrate the same point on a footprint basis directly: Walong has 524 detected buildings (119,779 m² total footprint area) — a genuinely large settlement, consistent with its role as an airstrip town — against a sample-wide median of 34; Kaho has 78 and Musai 108, both well above the median, consistent with both having received “model village” investment rather than being negligible hamlets. None of the three is an outlier in the sense of having so few buildings that a small VVP addition would move a percentage-based index dramatically, which is itself informative: Walong’s confirmed new terminal building and hostel are additions to an already-substantial built environment, exactly the kind of addition a 500m area-average is least likely to register as a large percentage change, and exactly what Section 4.10/7.3 found NDBI failed to register there.

7.3 Ground-Truth Positive-Control Validation — Completed (see Section 6.6)

An earlier version of this section reported that no known, dated, independently-verified completed VVP-I initiative could be identified that also fell inside the 258-village geocoding sample. That search has now been redone and succeeded (Development Log Entry 28). News coverage of an official two-day tour of Anjaw district, Arunachal Pradesh, by the state’s Deputy Chief Minister (Arunachal Times, arunachalobserver.org, 31 October – 1 November 2023) documents 15 specific projects inaugurated across several Anjaw villages, three of which are in this study’s 258-village sample: **Walong** (a civil terminal building at Walong’s advanced landing ground, and a 30-bed girls’ hostel at Walong Government Secondary School), **Kaho** (a basketball court, solar street lights, and a school library, explicitly reported as funded “under Vibrant Village Program”), and **Musai** (solar street lights and other “model village” facilities, also explicitly VVP-attributed). All three inaugurations fall inside this study’s 2021-2025 comparison window, with a specific date attached (31 October – 1 November 2023), which is exactly the “specific, dated, named” positive-control case the earlier version of this section said could not be found.

Checking each of this study’s four extracted proxies against these three villages individually (all values full-year, 2021 before vs. 2025 after, from the same extractions Sections 4.2, 4.3, and 4.10 already use): VIIRS night-lights shows a large, clearly-detected increase at all three — Walong 0.892 → 1.801 (+0.909, essentially doubling), Kaho 0.514 → 0.934 (+0.420, an 82% rise), Musai 0.317 → 0.328 (+0.011, small but still positive) — Walong and Kaho’s jumps are well above this study’s own pooled full-year lights mean change, consistent with the confirmed addition of new lighting infrastructure. NDBI, by contrast, moves in the *opposite* direction at all three villages despite the confirmed new construction: Kaho -0.057, Musai -0.064, Walong -0.038. Dynamic World’s “built” probability (Section 4.10) shows a small increase at all three (Kaho +0.044, Walong +0.016, Musai +0.006) — smaller than night-lights’ jump but at least directionally consistent with real development, unlike NDBI. SAR VV/VH (Section 4.10) shows no clear signal either way at any of the three (changes under 0.15 dB, within the noise this study’s pooled SAR result is already null at).

This is a 3-village case study, not a statistical test, and it should not be read as more than that: it cannot establish that VIIRS is “the correct” proxy or that NDBI is “wrong” in general, and three inaugurations in one official visit to one district is not a random sample of VVP-I’s investment portfolio across five states. But within those limits, it is real evidence, not conjecture, for a specific and useful explanation of part of this study’s own null: the confirmed projects here (solar streetlights, a hostel, a basketball court, a terminal building) are all small, point-scale additions, exactly the “individual roads, buildings, small installations” Section 7.2 already flagged NDBI’s 500m area-average as poorly suited to detect — a few new light fixtures shift a radiance sensor’s reading directly, but the same fixtures are a rounding error inside a 500m-radius reflectance average dominated by terrain, vegetation, and existing built surface. Read together with Section 7.2’s own proposal for sub-500m or building-footprint-level detection, this case study is a concrete reason to expect that approach would matter, not just a generic call for more resolution.

7.4 RTI Follow-Through for Himachal Pradesh and Ladakh

In fact, in this study, the purposeful decision was made not to press RTI requests for these two states’ missing village-wise annexures, which are recorded in the `BO_Development_Log.md`. That must be reversible — requesting those would fill in the two biggest known gaps in primary-source information coverage instead of the illustrative or excluded treatment as documented here.

7.5 A Genuine Pre-Treatment Panel for the Control Group — Completed, Mostly Reassuring, One Flagged Exception

A true pre-treatment panel did not exist because this study’s control-group comparison (Sections 3.7, 4.6, 6.7) only ever had one pre-period (2021) to work with — the level-balance check in Section 6.7 could confirm treated and control villages started at similar *levels* in 2021, but not that they were moving in parallel *before* treatment, which is what the DiD design’s own identifying assumption actually requires. That gap is closed (Development Log Entry 30): a second, earlier pre-treatment year (2019, both windows, both groups) was extracted, and a placebo DiD was run comparing the 2019-to-2021 change between treated and control villages — a period during which VVP-I could not possibly have had an effect, since the programme was not sanctioned until February 2023 (Section 1). If treated and control villages already showed a significantly different trend before treatment could exist, that would undercut the real 2021-to-2025 DiD’s parallel-trends assumption; a null placebo result is the reassuring outcome, not a disappointing one.

Three of the four checks are clean: full-year NDBI (placebo DiD coefficient +0.00196, $p=0.864$), full-year lights (-0.00665, $p=0.755$), and summer lights (+0.00047, $p=0.968$) all show no significant pre-treatment divergence between treated and control villages — direct evidence *for*, not merely an assumption behind, this study’s DiD design in three of its four outcome/window combinations. The fourth is a genuine, flagged exception: summer NDBI’s placebo DiD coefficient is +0.01287, $p=0.064$ — not significant at the conventional threshold under the district-fixed-effects specification, but close to it, and the unadjusted Mann-Whitney comparison of the raw treated-vs-control distributions for this same combination is sharply significant ($p=0.00001$). Read together, this says district composition explains most, but evidently not quite all, of a real pre-existing difference in how treated and control villages’ summer NDBI moved between 2019 and 2021 — a borderline result, not a clean pass, and it should be

reported as such rather than rounded up to “passes” because the district-adjusted p-value happens to clear 0.05. It is worth noting the borderline pre-period effect (+0.01287, treated trending up) runs in the *opposite* direction from the real, already-null summer NDBI DiD (-0.00357, Section 4.6) — so this is not a case of a pre-existing trend mechanically continuing into and explaining away the main result; if anything a naive reader might have expected the pre-trend to inflate the real effect, and it does not appear to have.

Taken as a whole, this is the strongest evidence this study has had for its own parallel-trends assumption — three of four combinations pass cleanly, and even the one exception does not point in a direction that would explain this study’s own (already null) headline finding. It does not fully retire the concern Section 6.7 raised, since one borderline result on one important pre-condition check is not nothing, but it moves this study’s own most consequential unaddressed threat (as this section previously described it) from “untested” to “mostly tested and largely reassuring, with one specific, honestly-reported exception.”

7.6 Stress-Testing the New Summer H3 Border-Proximity Result — All Three Checks Now Done

Section 4.5’s summer-window finding that built-up change correlates with distance to the border/LAC ($\rho = 0.291$, $p = 0.0000026$) emerged only once this paper’s own extraction was completed (Development Log Entry 22). It has now been put through all three checks the previous, since-retracted summer NDBI-change result was built around, not two: leave-one-district-out

(src/analysis/h3_border_proximity_robustness.py) reruns significant in 14 of 14 district-dropped subsamples, same sign throughout, ρ ranging [+0.183, +0.337] — no single district is driving it — and randomization inference (2,000 permutations of the outcome against distance) puts the observed ρ beyond every one of the 2,000 shuffled values ($p < 0.001$). Unlike the summer NDBI-change result and the full-year lights DiD, both of which failed this exact pairing of checks once actually run, this result survives it. The same script also ran this pairing on the *other* Holm-significant H3 result, full-year night-lights-proximity ($\rho = -0.252$) — described elsewhere in this paper (§4.5, §4.9) as robust because it has stayed significant “as it always has” across several data re-extractions, which is a different and weaker claim than actually having been leave-one-out or randomization-inference checked. It was not, until now: it also survives (14/14 leave-one-out, $p < 0.001$ randomization inference, ρ range [-0.292, -0.175]).

The third item — linearity across the full distance range versus concentration at one end — and a direct comparison against the three-point 2021/2023/2025 extraction (Section 3.8) are both now done as well (Development Log Entry 27), and the results are more mixed than the leave-one-out/randomization pairing above. On linearity: the summer NDBI-proximity correlation is close to linear across all five distance quintiles (nearest to farthest: mean change -0.0182, -0.0074, +0.0009, +0.0071, +0.0135 — a genuinely progressive gradient, quadratic term not significant, $F=0.89$, $p=0.347$) — a clean result on this check. The full-year lights-proximity correlation is not: the same five bins give +0.0815, +0.1557, +0.0090, +0.0073, +0.1765 — rising, falling twice, then jumping back up in the farthest bin, not a monotonic gradient, even though a formal quadratic test still does not flag it as curved ($F=0.69$, $p=0.406$). The honest read is that this correlation’s overall shape looks more consistent with a small number of districts happening to sit at particular distances than with a smooth proximity effect — a real caveat specific to the full-year lights result, on top of (not replacing) its leave-one-out/randomization robustness above.

On the multi-year cross-check: both correlations reproduce closely against the independent three-point extraction over the full 2021-2025 span (summer NDBI $\rho=+0.293$ versus the two-point extraction's own $+0.291$; full-year lights $\rho=-0.252$ versus -0.252 exactly) — a genuine independent-data consistency check, since Section 3.8's extraction is a separate pull from the two-point comparison. Split into the two sub-periods, though, neither correlation is temporally stable. Summer NDBI's relationship actually reverses sign: distance vs. 2021-2023 change is $\rho=+0.426$ ($p<0.00001$, same direction as the headline result, stronger), but distance vs. 2023-2025 change is $\rho=-0.222$ ($p=0.0004$, opposite sign, also significant) — the reported positive correlation over the full period is a net of two oppositely-signed halves, not a steady gradient holding throughout, which fits with Section 4.7's own finding that the pooled two-point NDBI comparison hides a decline-then-recovery pattern. Full-year lights is more consistent but still uneven: 2021-2023 $\rho=-0.239$ ($p=0.0001$, same direction) weakens to 2023-2025 $\rho=-0.078$ ($p=0.218$, same direction, no longer significant) — concentrated in the earlier half of the study period rather than absent from either half.

Neither of these newer findings overturns either H3 result — both remain genuine, Holm-significant correlations on the actual 2021-2025 comparison this study specifies, and both still pass the leave-one-out and randomization-inference pairing above. But “stress-tested and surviving” now needs a more precise description than it had before this entry: surviving a district-dropout and a label-permutation check is not the same as being a stable, linear, time-consistent relationship, and on that finer reading only the summer NDBI-proximity result's linearity holds up cleanly — its temporal stability does not, and neither does the full-year lights-proximity result's linearity. Section 6.11 adds a further, related caveat that applies to both: neither correlation's reported p-value yet accounts for spatial autocorrelation among nearby villages, which this study has now confirmed is real and significant.

7.7 Statistical Power: What This Design Could Have Detected

A null result answers “was there a significant effect”; it says nothing on its own about whether this study's design could have detected a real effect if one existed. `src/analysis/power_analysis.py` computes, for the primary H1 (treated-only, paired) and H4 (control-group DiD) tests, the Minimum Detectable Effect (MDE) at conventional significance ($\alpha=0.05$, two-sided) and power (80%), using t-quantiles at this study's own degrees of freedom ($n-1=250$ for H1; 14 districts – 1 = 13 for H4, per Cameron & Miller's (2015) guidance for a small-cluster-count design) rather than a large-sample z approximation.

H1's MDE, expressed as Cohen's d, is 0.178 in every outcome/window combination — a design property that depends only on n and df, not on the outcome's own scale — which sits just under Cohen's conventional 0.2 threshold for a “small” effect. In plain terms: this study's primary design was sensitive enough to have caught a small-to-moderate real effect had one existed, so H1's null is a substantive finding, not an artifact of insufficient power. H4's cluster-based design is, as expected, less sensitive (MDE of 0.01992/0.15045 raw units for full-year NDBI/lights, 0.01505/0.13017 for summer), but every observed H4 coefficient sits well inside that threshold too (24-34% of the respective MDE) — not a near-miss, a clear gap between what was observed and what the design could have detected.

8. Conclusion

This study cannot conclude that the pace of development a budget of this size would imply, let alone the steady annual progress trajectory the scheme aims toward, has been achieved at the priority villages within India's Vibrant Villages Programme in five border states and union territories. On its primary measure — NDBI and night-lights, QA60-masked, against the treated-only comparison, against the control group, at any buffer radius tested, under an exact wild-cluster bootstrap addressing this study's small (14) cluster count, and under a genuine 2019-to-2021 parallel-pre-trends placebo test — this study cannot conclude that a satellite-detectable effect exists. This is the settled version of a conclusion this study reached twice, each time by finding and fixing a real problem rather than by assuming the answer. First: an earlier version reported a significant summer-window built-up-area increase that survived every robustness check run against it; that result depended on an incomplete extraction missing all 31 of Sikkim's villages, and once that extraction was completed it reversed to a clean null (Sections 4.2, 4.6, 4.8; Development Log Entries 21–22). Second: the full-year night-lights control-group gap that took that result's place was itself found to rest on a contaminated control group, and once fixed, went null too on every specification (Development Log Entries 23–25).

Two things now sit alongside that null, and they are not the same kind of thing as each other. A border-proximity correlation this paper once flagged as new and unchecked has since been run through leave-one-district-out, randomization inference, and a spatial-autocorrelation-corrected permutation test, and survives all three (Sections 6.11, 7.6) — a real secondary finding, still correlational, running opposite to what securitization theory predicts. And triangulating the primary measure against two satellite proxies did not simply confirm the null: Sentinel-1 SAR (the genuinely sensor-independent one) agrees, but Dynamic World's built-up classifier shows a significant control-group effect in both windows and validates more strongly against real building-footprint ground truth than NDBI does, while directly missing all three of this study's confirmed real-construction case studies that Dynamic World and night-lights correctly detected (Sections 4.10, 6.6, 7.2, 7.3) — this one is a genuine, unresolved tension, not a bug. A third item that looked like a second open tension in an earlier version of this section — the QA60-vs-SCL cloud-mask cross-check appearing to flip the summer NDBI result from null to significant — turned out on external review to be a coding bug in the SCL mask itself (snow pixels counted as "clear"); fixed and rerun, the two masks now agree, both null (Section 6.12, Development Log Entries 33–34). That third item is reported here as a resolved bug, not a third open finding, and readers should weigh it accordingly: it says something about this project's own review process catching an error rather than about the underlying science being unsettled.

None of this reverses the primary finding the way the two earlier artifacts did (Sikkim's missing imagery, the contaminated control group); each of those was found to be a data problem and, once fixed, disappeared entirely, and the cloud-mask bug just described is a third instance of exactly that pattern — a coding error that looked like a substantive finding until it was fixed. What triangulation surfaces, by contrast, is not a bug: this study's own primary measurement approach — one satellite index — is not the only reasonable way to ask this question, and at least one other reasonable way (a different built-up classifier) does not agree with it, and that disagreement has not been explained away by any check run against it so far. Budget and measured change still show no comparable relationship in either window, and what little pattern three states can show flips sign between compositing windows. Given the scale and strategic framing of this scheme, it should not continue without the independent impact assessment that its own parliamentary record confirms has never been conducted — and this study is itself now a demonstration of why that assessment needs to be done carefully, with more than one

measurement convention, and with its own checks re-verified rather than trusted on first pass: an incomplete extraction, a contaminated control group, and a snow-contaminated cloud mask each independently produced a result that read as robust until the underlying problem was actually found and fixed; a different proxy independently produced a result that disagrees with this study's own primary finding and, unlike the other three, has not been explained away. Any future assessment, official or independent, should be held to the same standard this study applied to itself repeatedly — complete, decontaminated data, pulled on a consistent timeline, checked against a control group, a buffer-radius sweep, a multi-year comparison, more than one satellite proxy, and more than one cloud-masking convention, each one actually re-verified when a check like external review flags a reason to doubt it — rather than whichever single-window, single-extraction, single-proxy result happens to be available first.

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